

PROGRAMMATION

Les structures

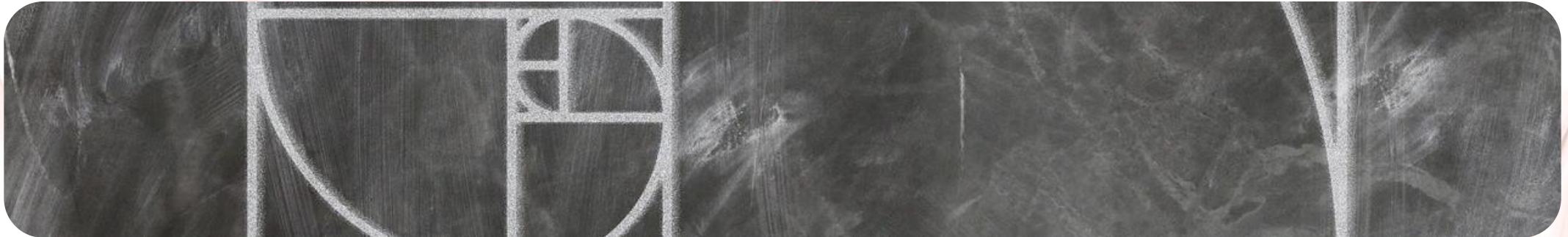
INTRODUCTION

Programmation - Les structures

- Les structures, comme leur nom l'indique, permettent d'agencer et de structurer un programme informatique de manière logique.
- Il en existe 3 types:
 - Les structures séquentielles
 - Les structures conditionnelles
 - les structures itératives

SÉQUENCES

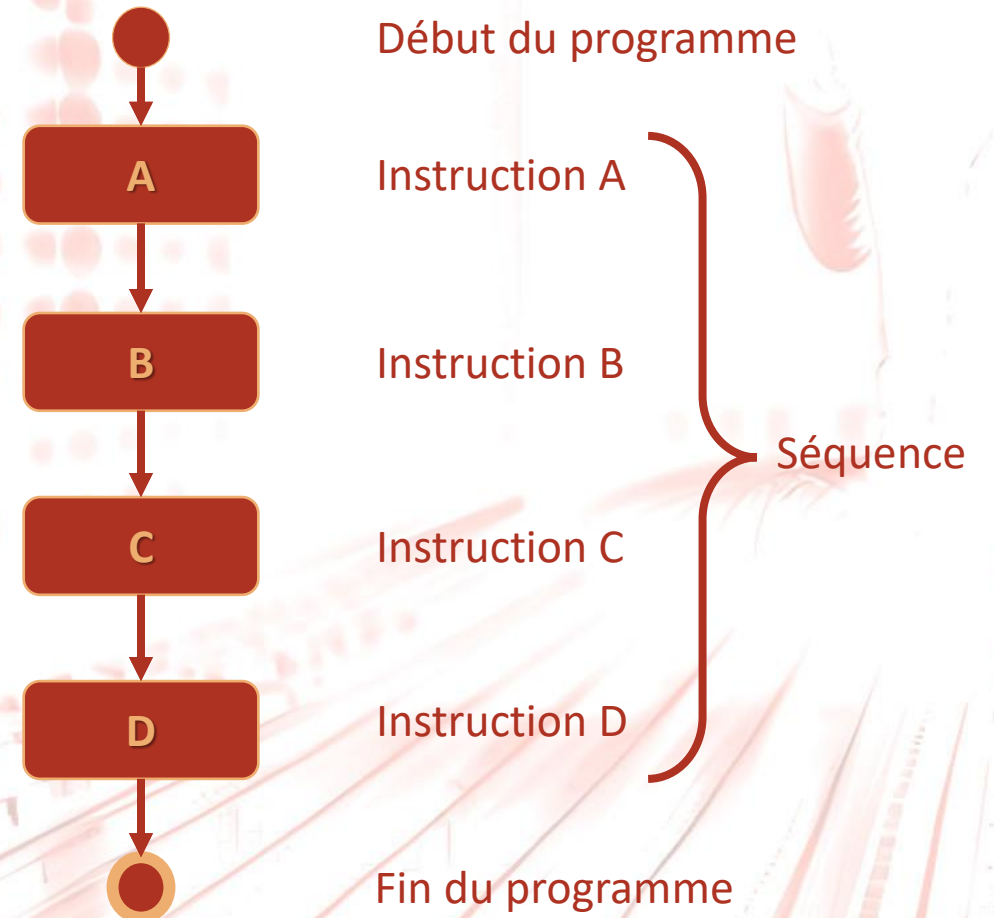
Programmation - Les structures



SÉQUENCES

Programmation - Les structures

- Les séquences sont des suites d'instructions séquentielles.
- Cela veut simplement dire que les instructions sont exécutées les unes à la suite des autres.

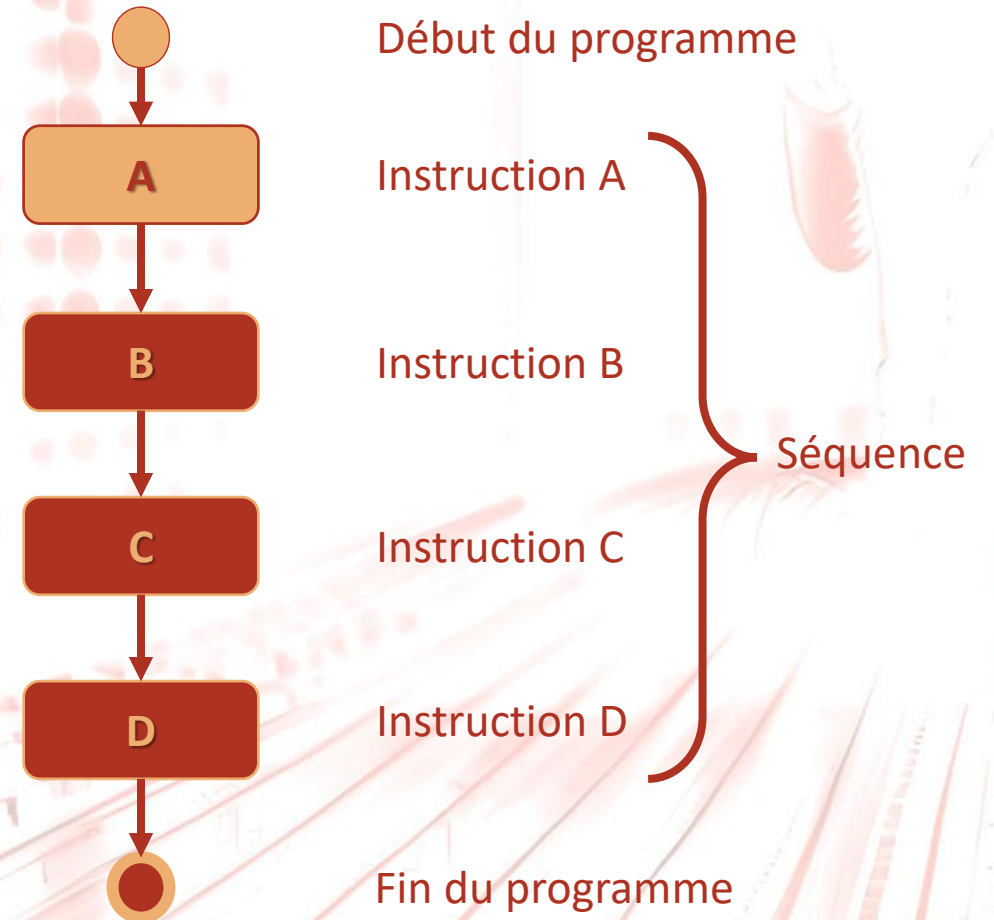


SÉQUENCES

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main() {
    cout<<"bonjour!"<<endl;
    cout<<"au revoir:"<<endl;
    return 0;
}
```

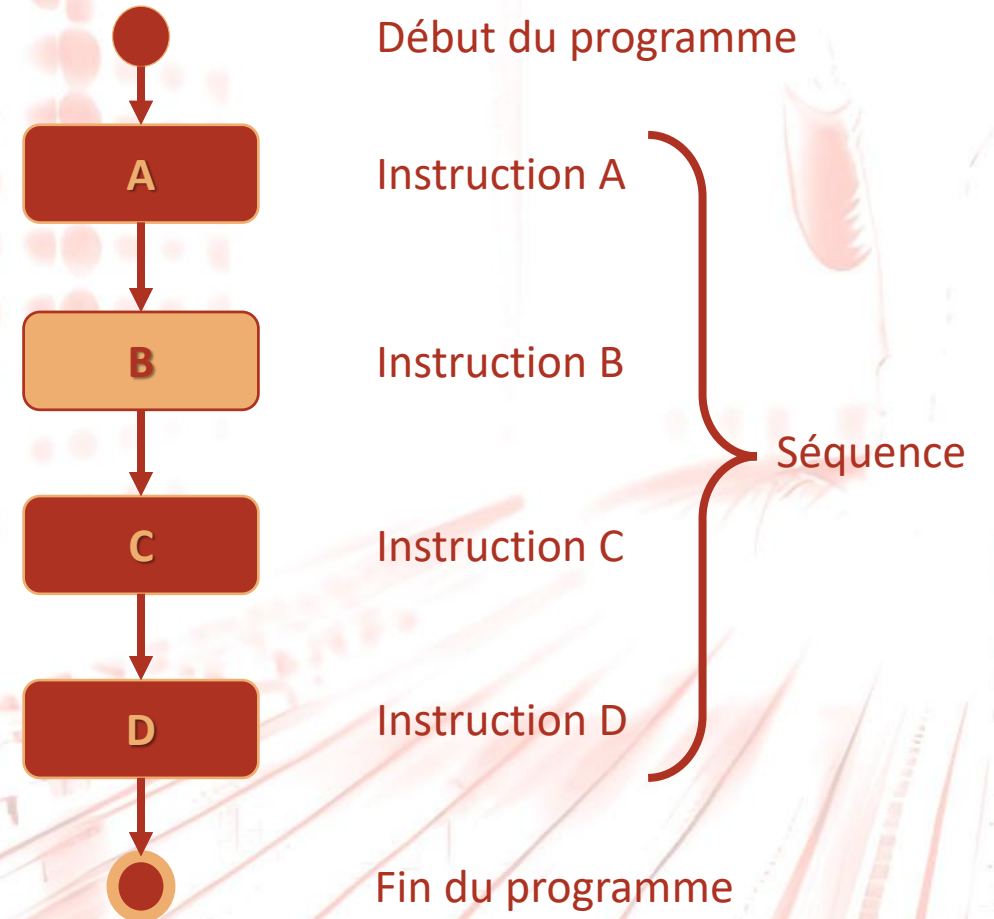


SÉQUENCES

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main() {
    cout<<"bonjour!"<<endl;
    cout<<"au revoir:"<<endl;
    return 0;
}
```

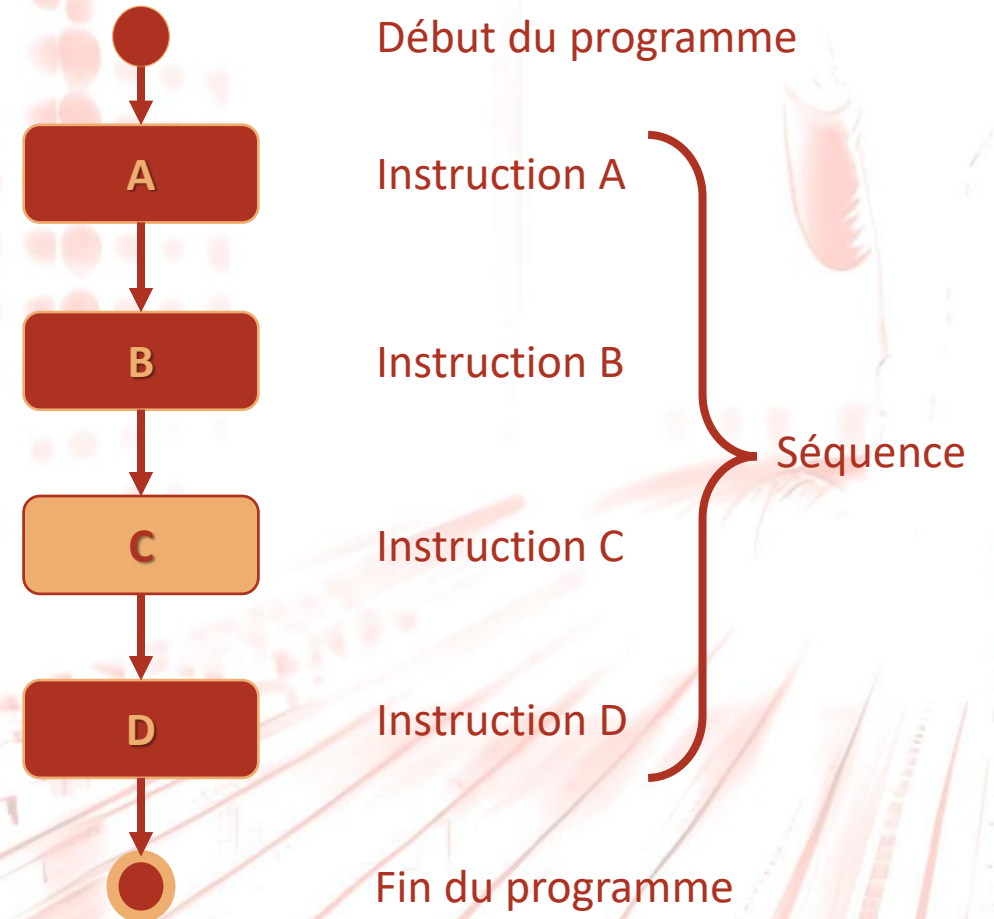


SÉQUENCES

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main() {
    cout<<"bonjour!"<<endl;
    cout<<"au revoir:"<<endl;
    return 0;
}
```

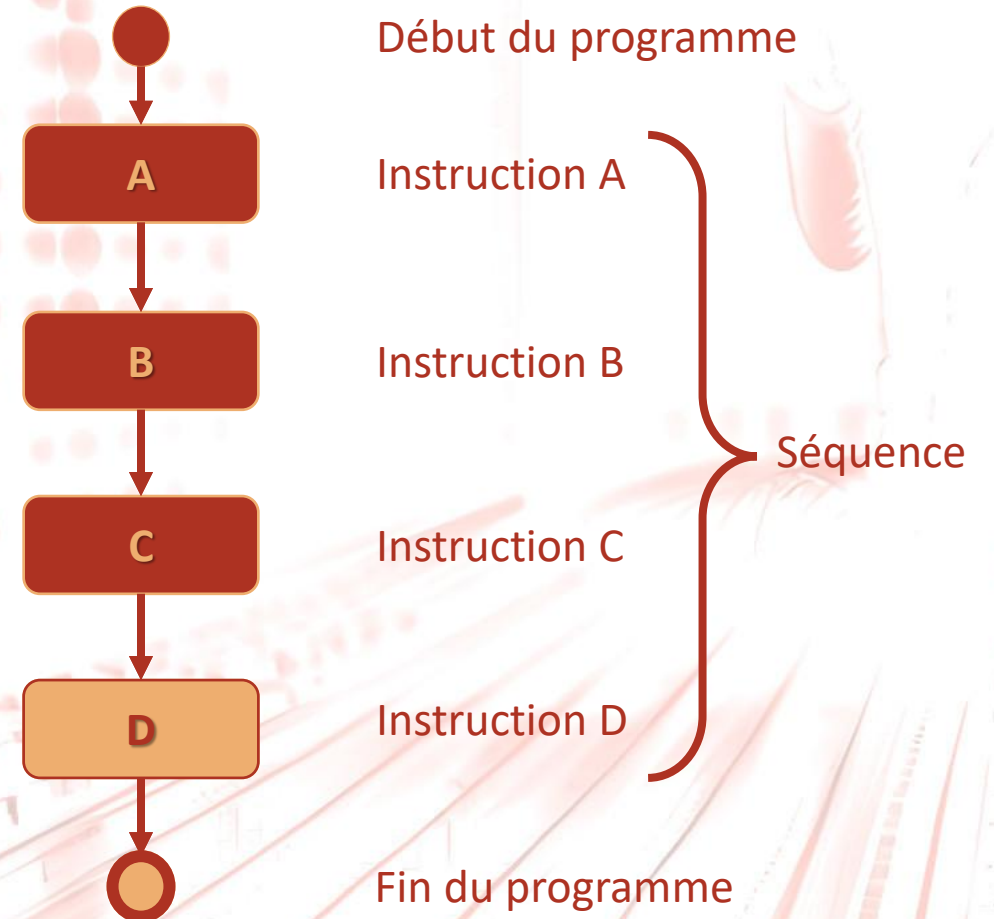


SÉQUENCES

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main() {
    cout<<"bonjour!"<<endl;
    cout<<"au revoir:"<<endl;
    return 0;
}
```



CONDITIONS

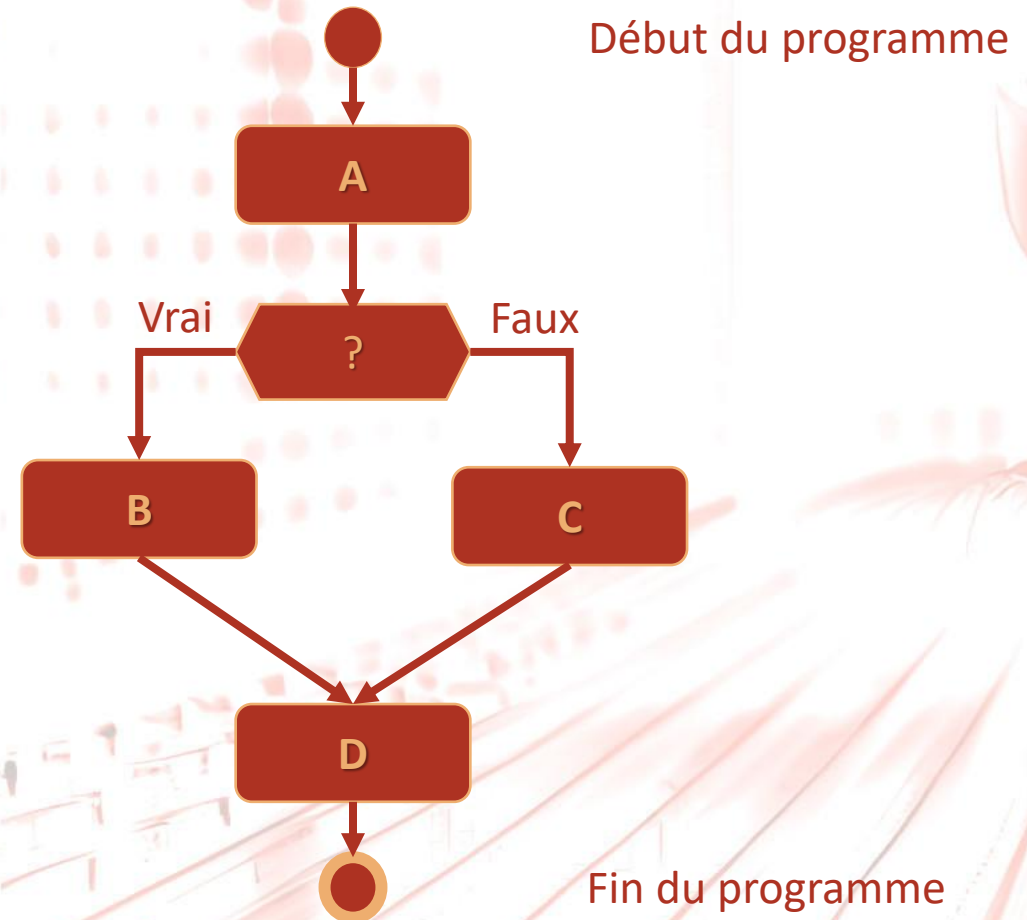
Programmation - Les structures



CONDITIONS

Programmation - Les structures

- Les conditions sont soumises à un test conditionnel.
- Suivant le résultat du test, on exécute une séquence parmi n
- En C++, il existe 2 structures:
 - le if...else
 - le switch case

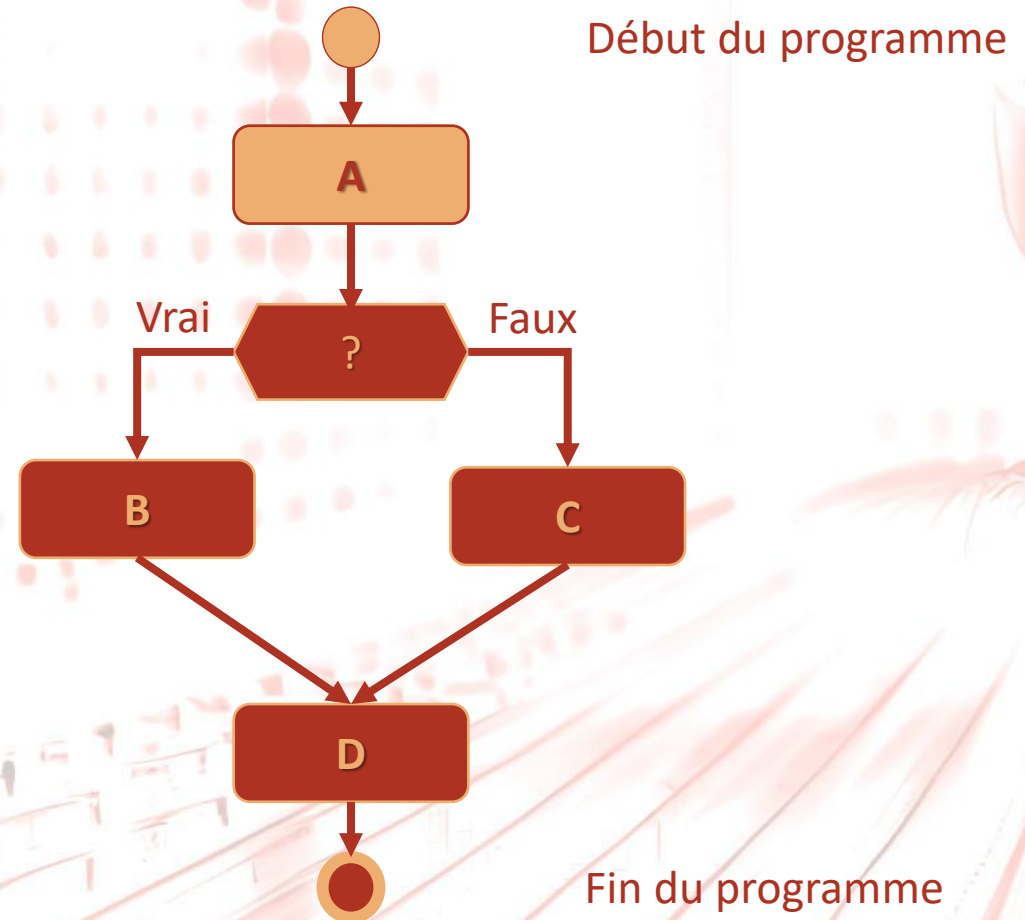


CONDITIONS: IF

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    if(nb == 0)
    {
        nb++;
    }
    else
    {
        nb--;
    }
    return 0;
}
```

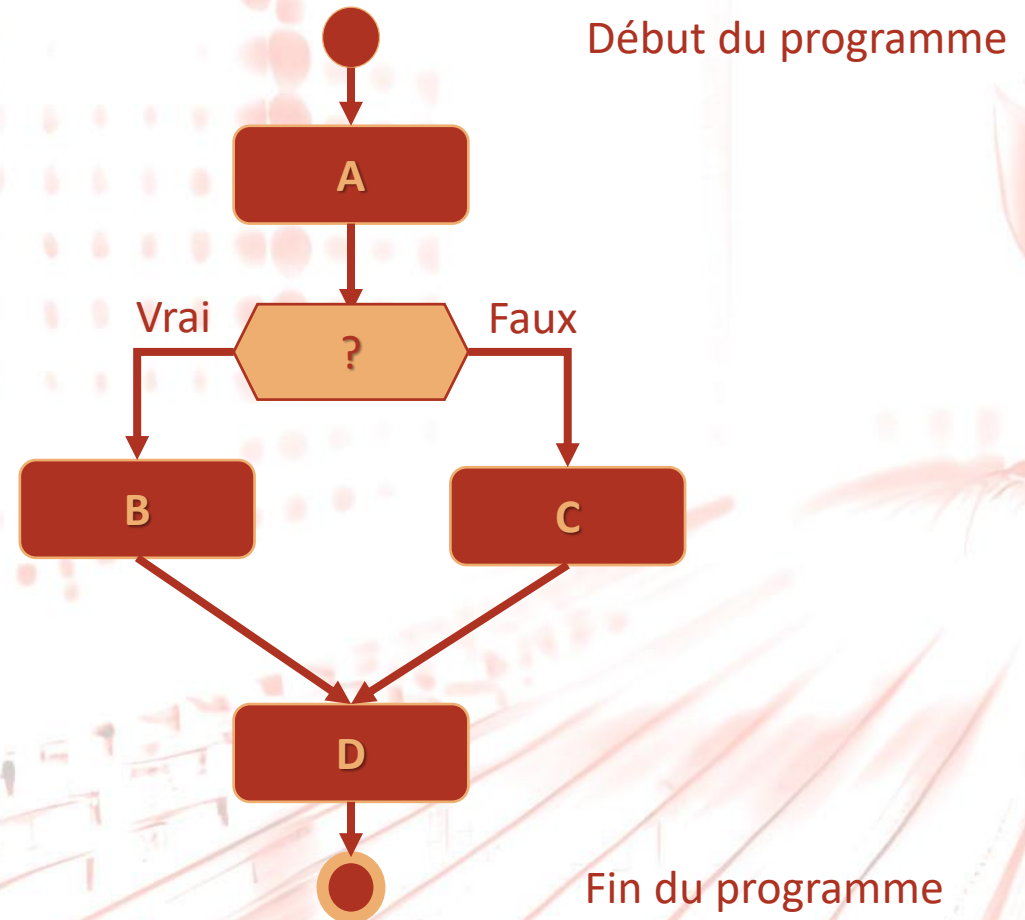


CONDITIONS: IF

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    if(nb == 0)
    {
        nb++;
    }
    else
    {
        nb--;
    }
    return 0;
}
```

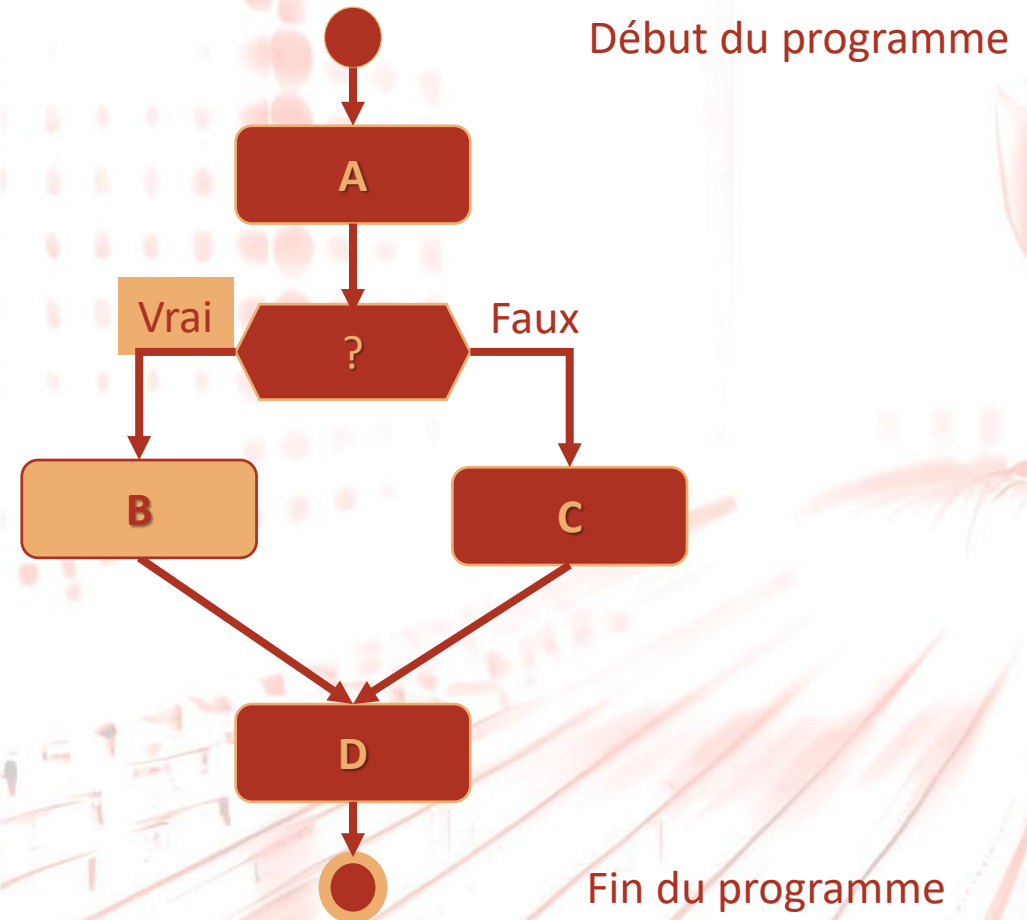


CONDITIONS: IF

Programmation - Les structures

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#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    if(nb == 0)
    {
        nb++;
    }
    else
    {
        nb--;
    }
    return 0;
}
```

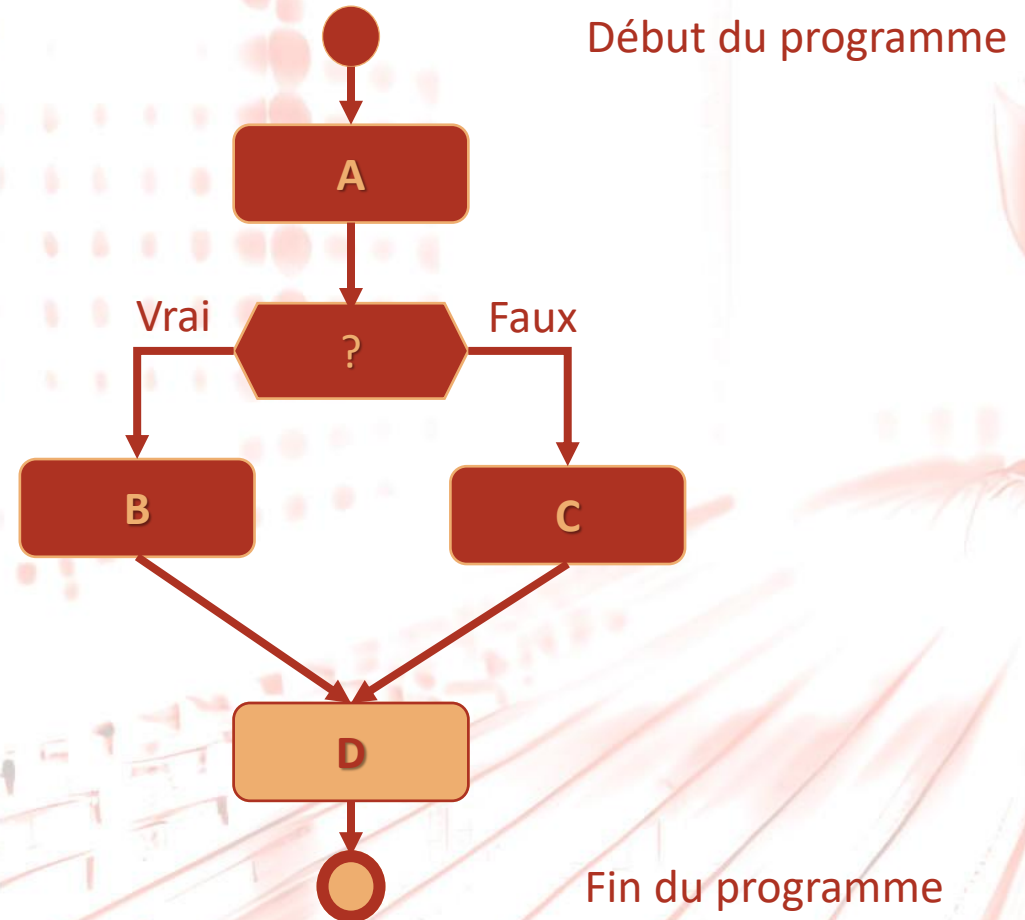


CONDITIONS: IF

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    if(nb == 0)
    {
        nb++;
    }
    else
    {
        nb--;
    }
    return 0;
}
```

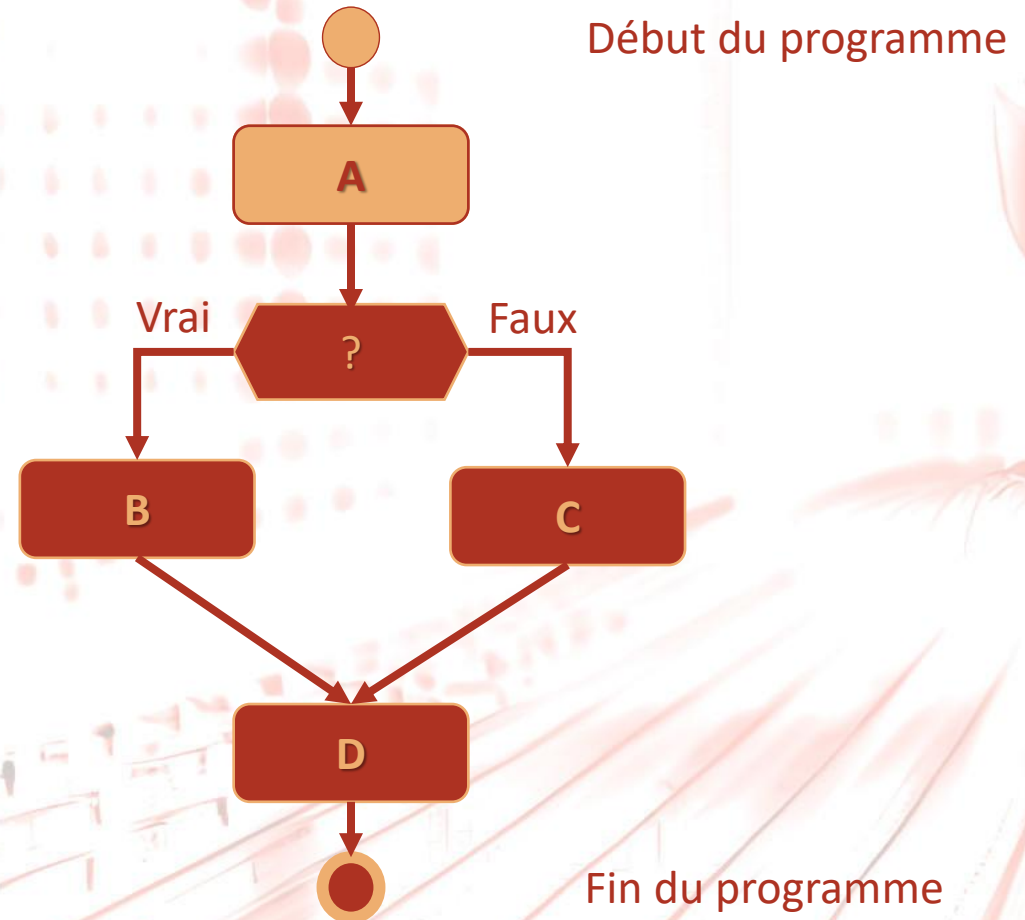


CONDITIONS: IF

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 1;
    if(nb == 0)
    {
        nb++;
    }
    else
    {
        nb--;
    }
    return 0;
}
```

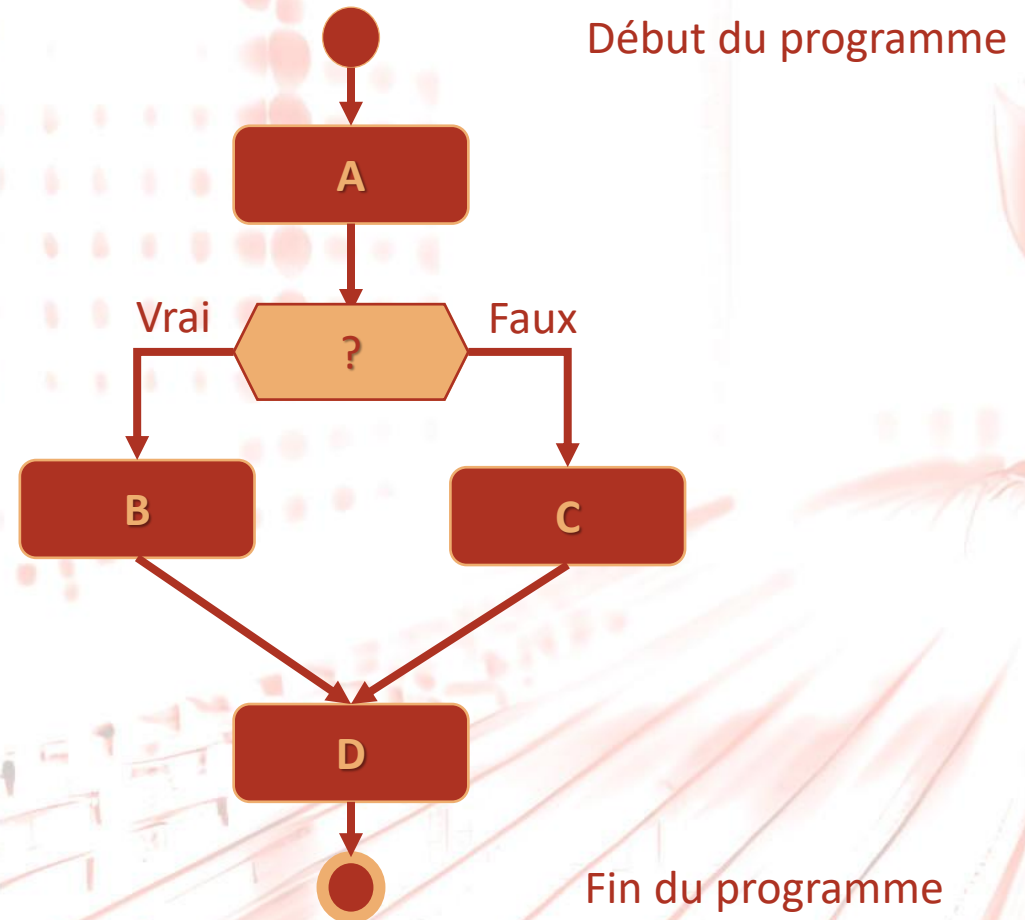


CONDITIONS: IF

Programmation - Les structures

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#include <iostream>
using namespace std;

int main(){
    int nb = 1;
    if(nb == 0)
    {
        nb++;
    }
    else
    {
        nb--;
    }
    return 0;
}
```

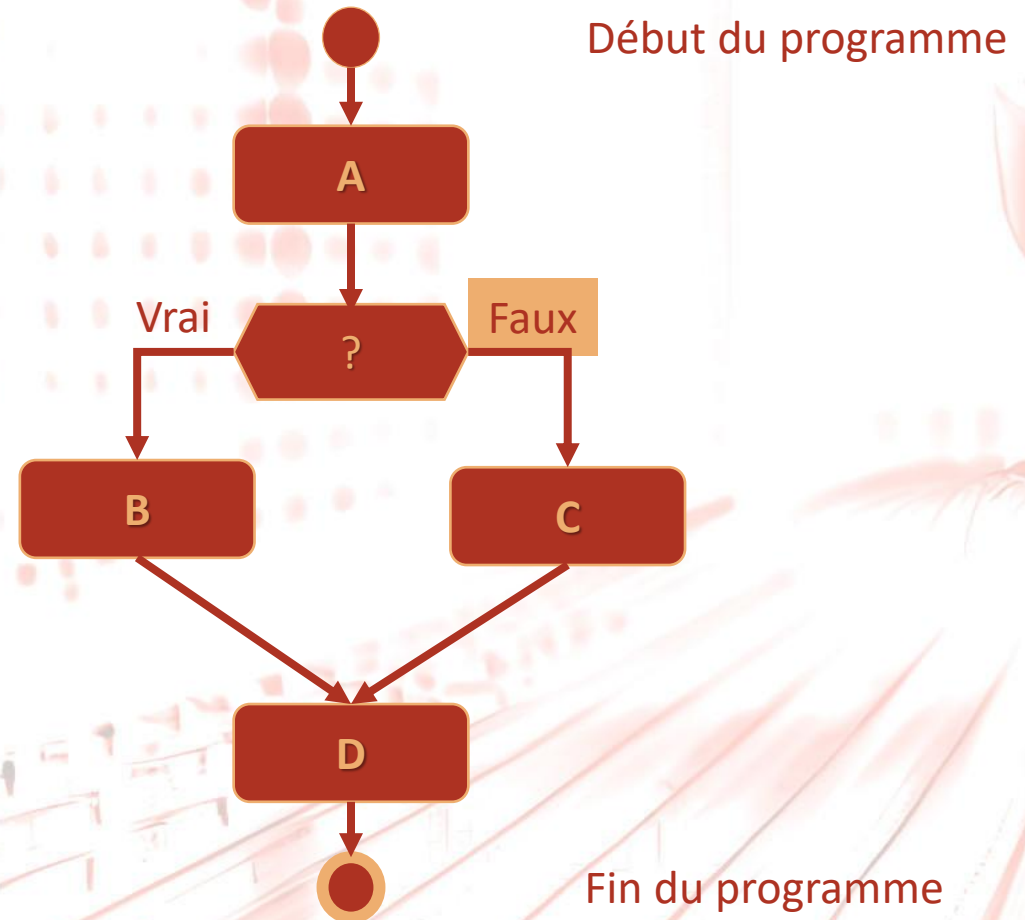


CONDITIONS: IF

Programmation - Les structures

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#include <iostream>
using namespace std;

int main(){
    int nb = 1;
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    {
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    }
    else
    {
        nb--;
    }
    return 0;
}
```

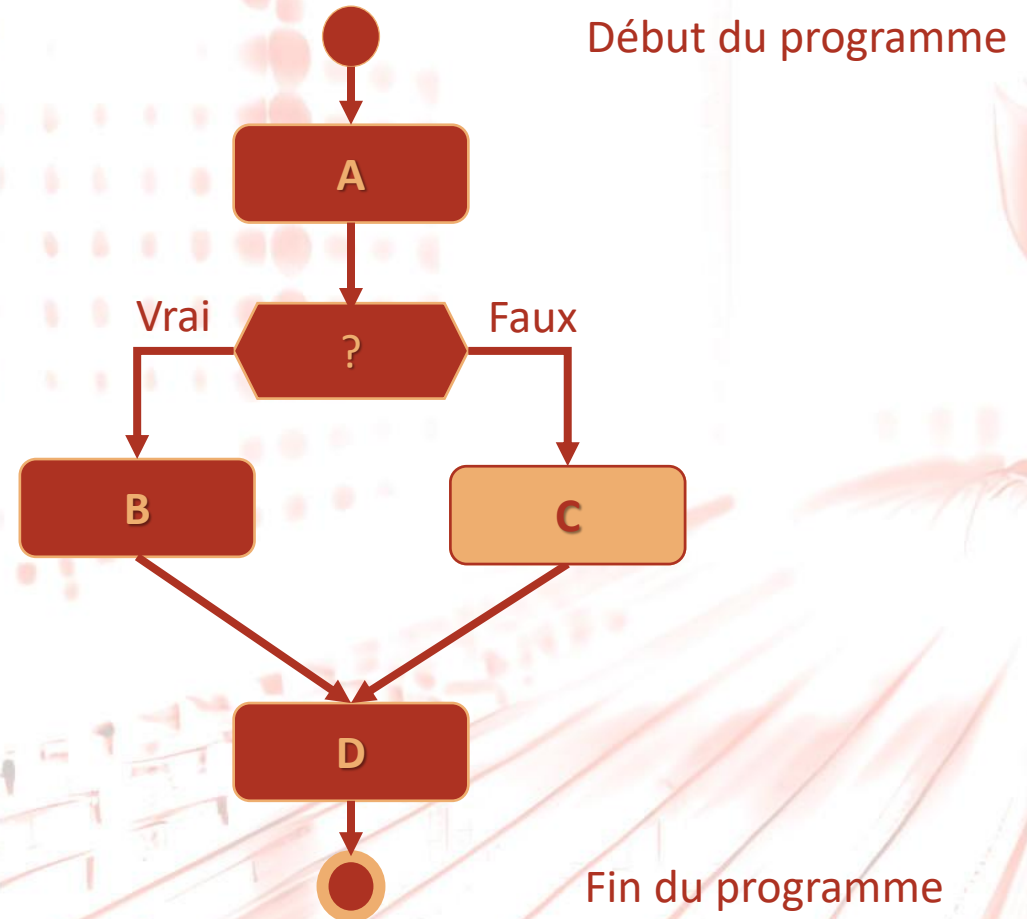


CONDITIONS: IF

Programmation - Les structures

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using namespace std;

int main(){
    int nb = 1;
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        nb++;
    }
    else
    {
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}
```

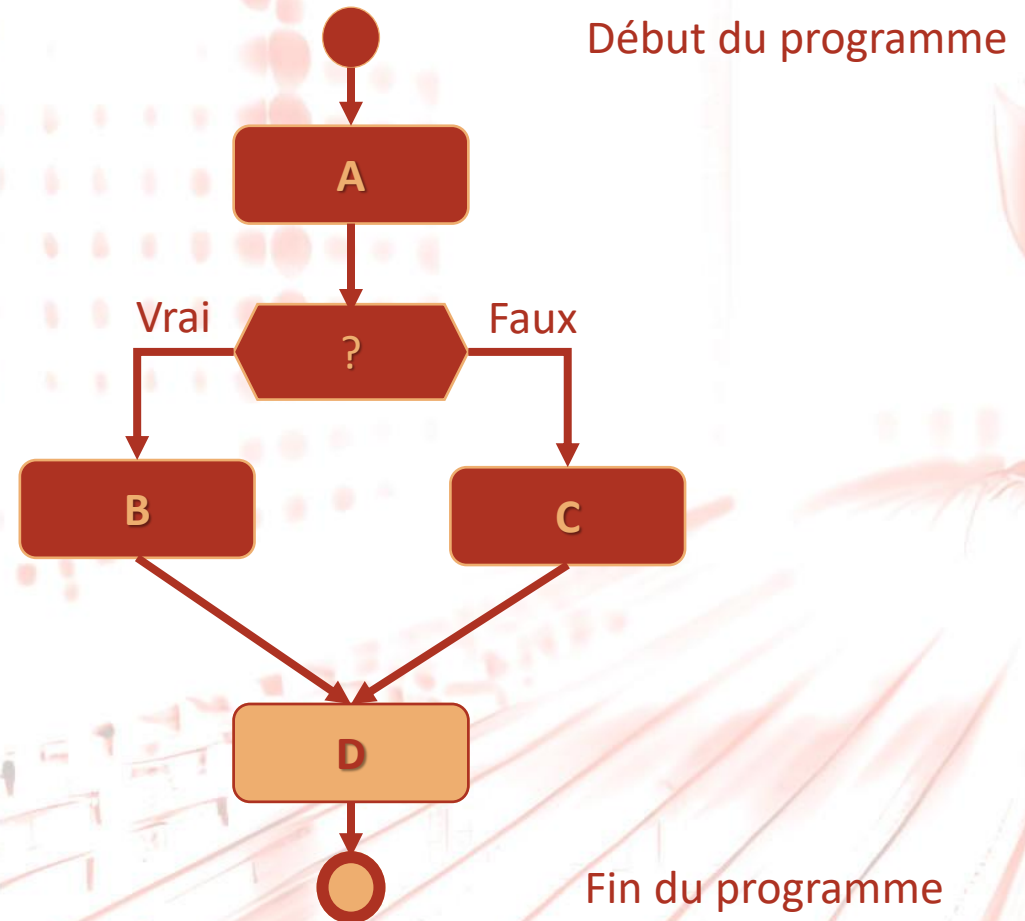


CONDITIONS: IF

Programmation - Les structures

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using namespace std;

int main(){
    int nb = 1;
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```

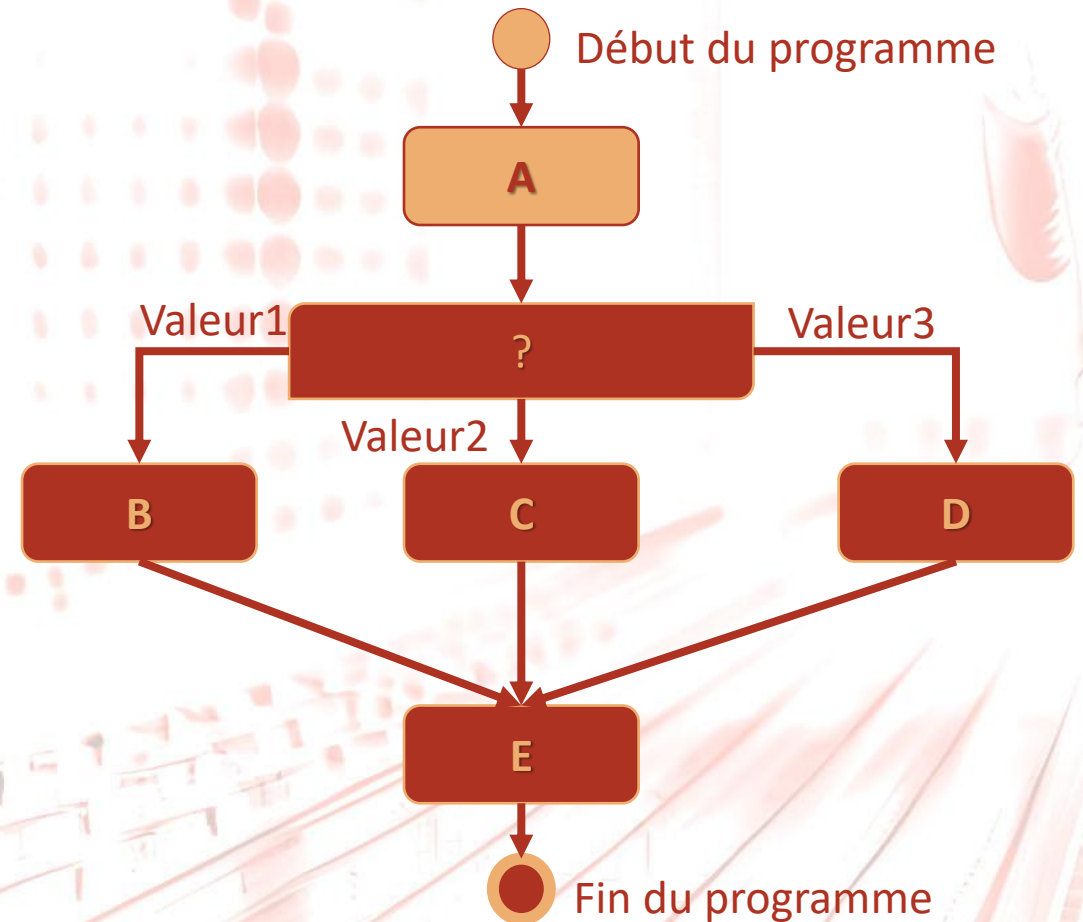


CONDITIONS: SWITCH CASE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main() {
    int nb = 0;
    switch (nb)
    {
        case 0: nb++; break;
        case 1: nb--; break;
        default: nb=3;
    }
    return 0;
}
```

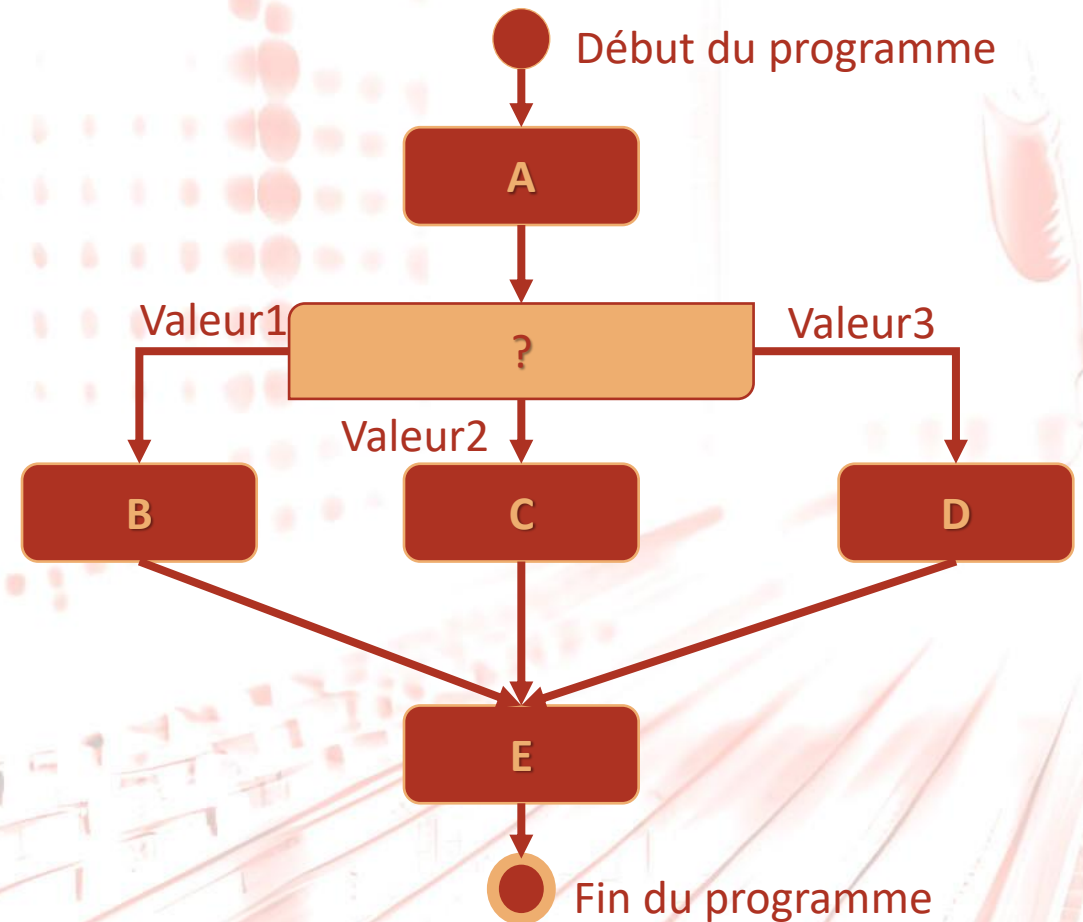


CONDITIONS: SWITCH CASE

Programmation - Les structures

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    return 0;
}
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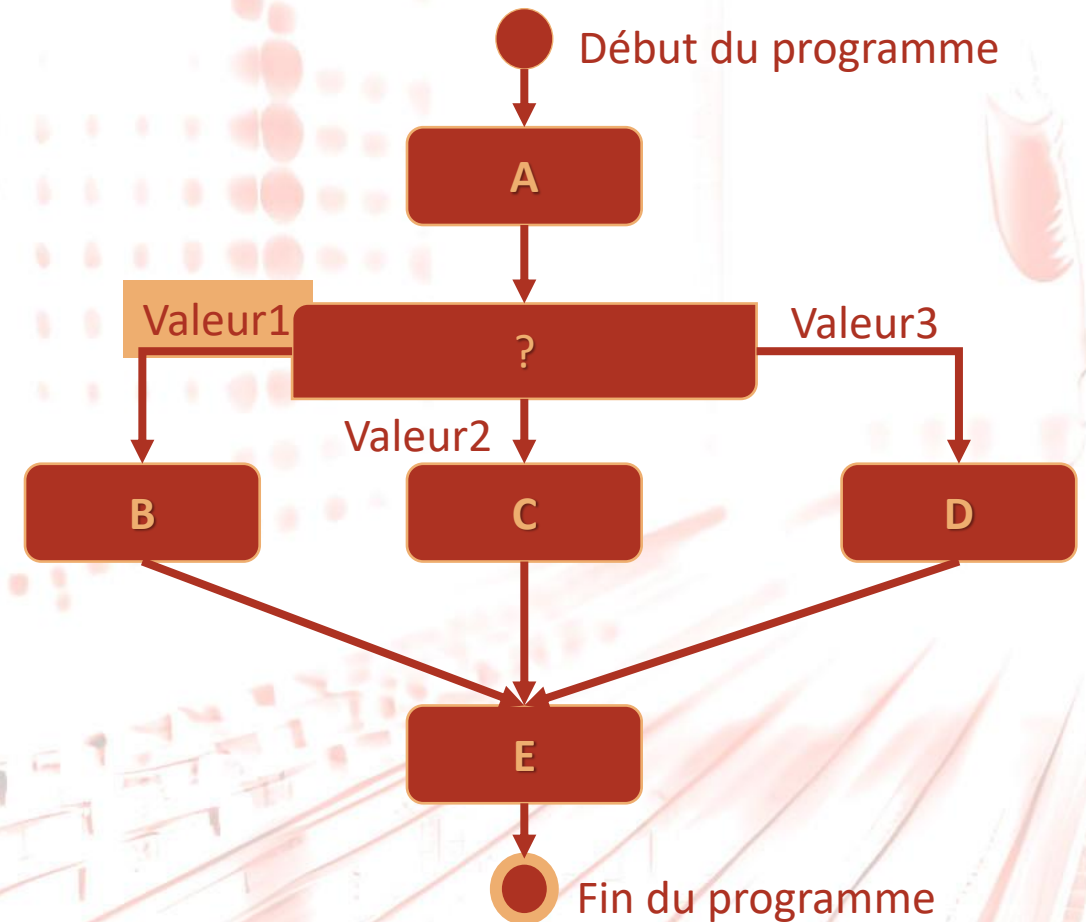


CONDITIONS: SWITCH CASE

Programmation - Les structures

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int main() {
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        default: nb=3;
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    return 0;
}
```

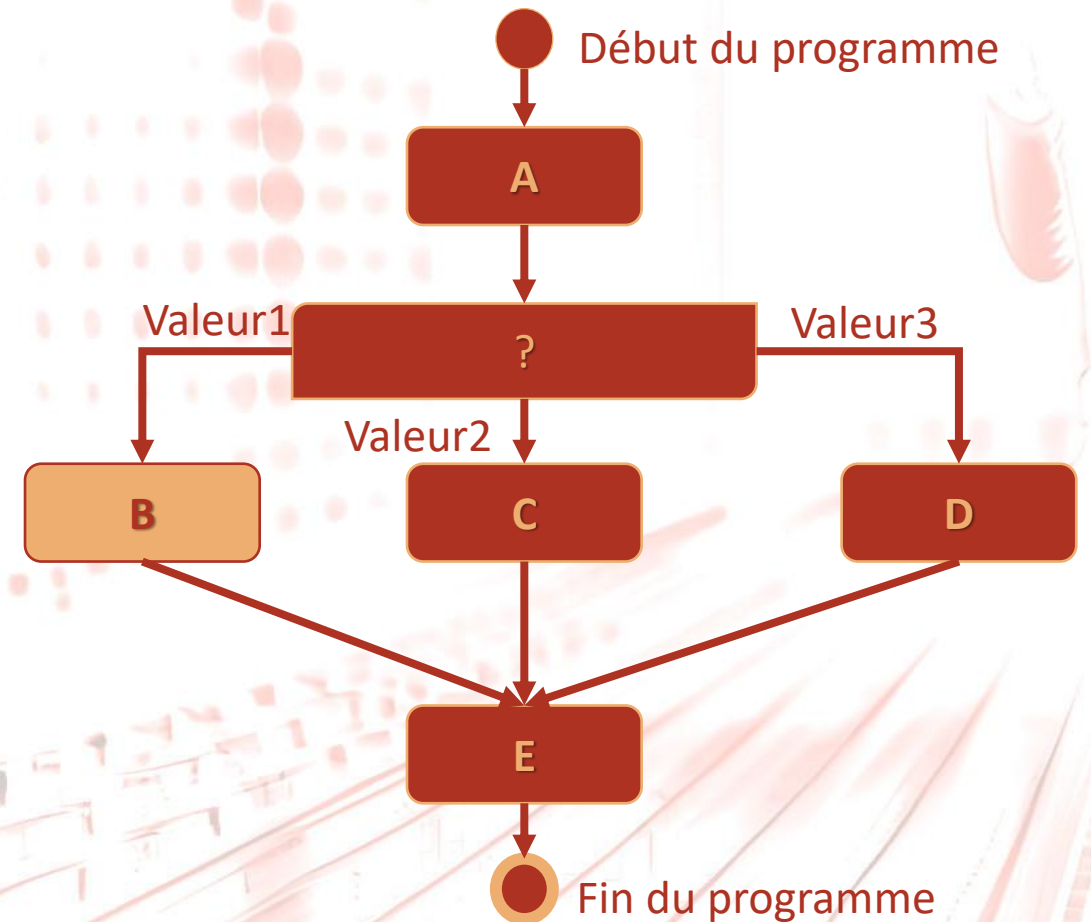


CONDITIONS: SWITCH CASE

Programmation - Les structures

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using namespace std;

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        default: nb=3;
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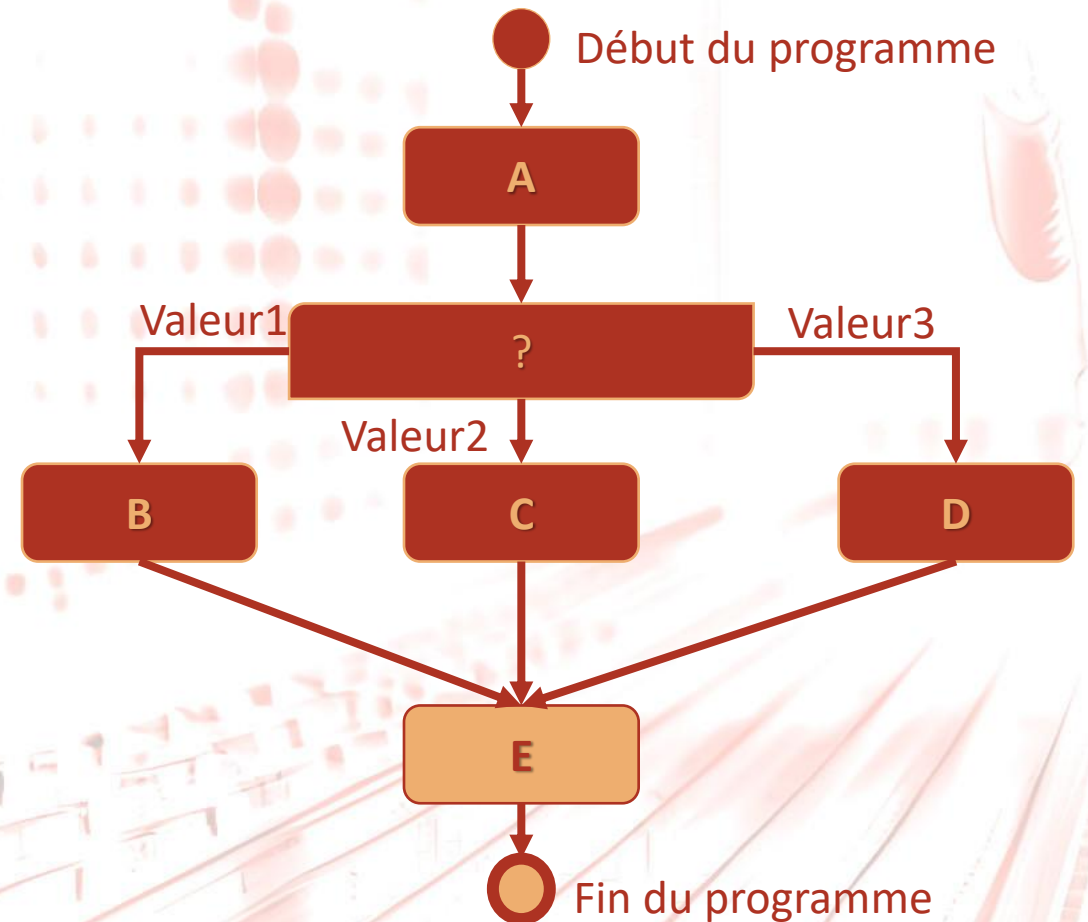


CONDITIONS: SWITCH CASE

Programmation - Les structures

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using namespace std;

int main() {
    int nb = 0;
    switch(nb)
    {
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        case 1: nb--; break;
        default: nb=3;
    }
    return 0;
}
```

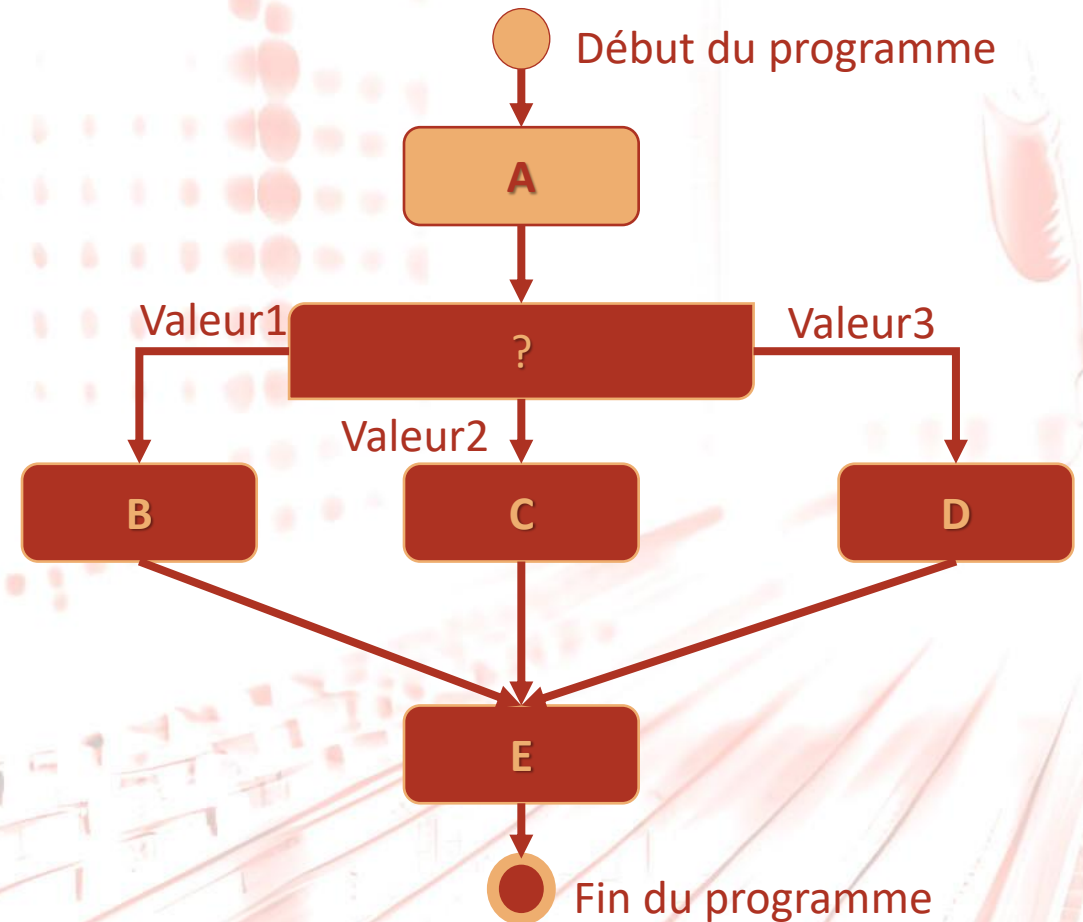


CONDITIONS: SWITCH CASE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main() {
    int nb = 4;
    switch (nb)
    {
        case 0: nb++; break;
        case 1: nb--; break;
        default: nb=3;
    }
    return 0;
}
```

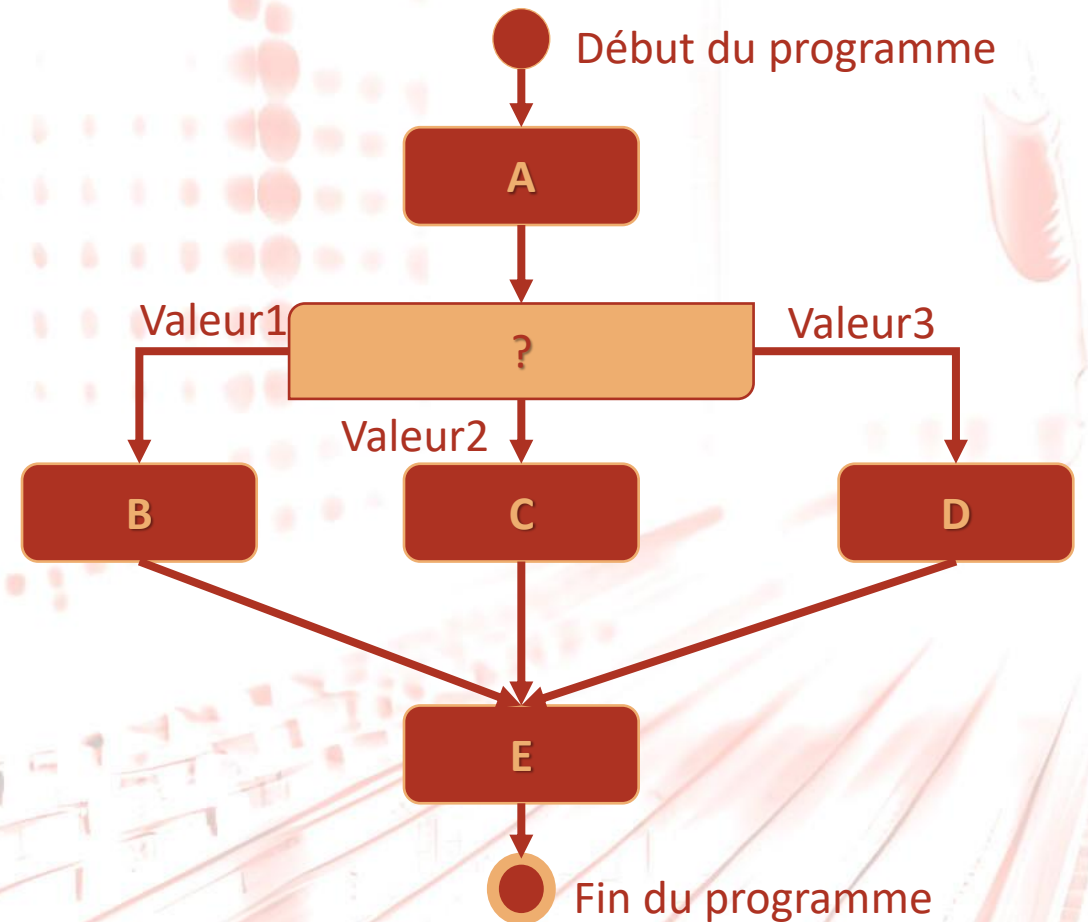


CONDITIONS: SWITCH CASE

Programmation - Les structures

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#include <iostream>
using namespace std;

int main(){
    int nb = 4;
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}
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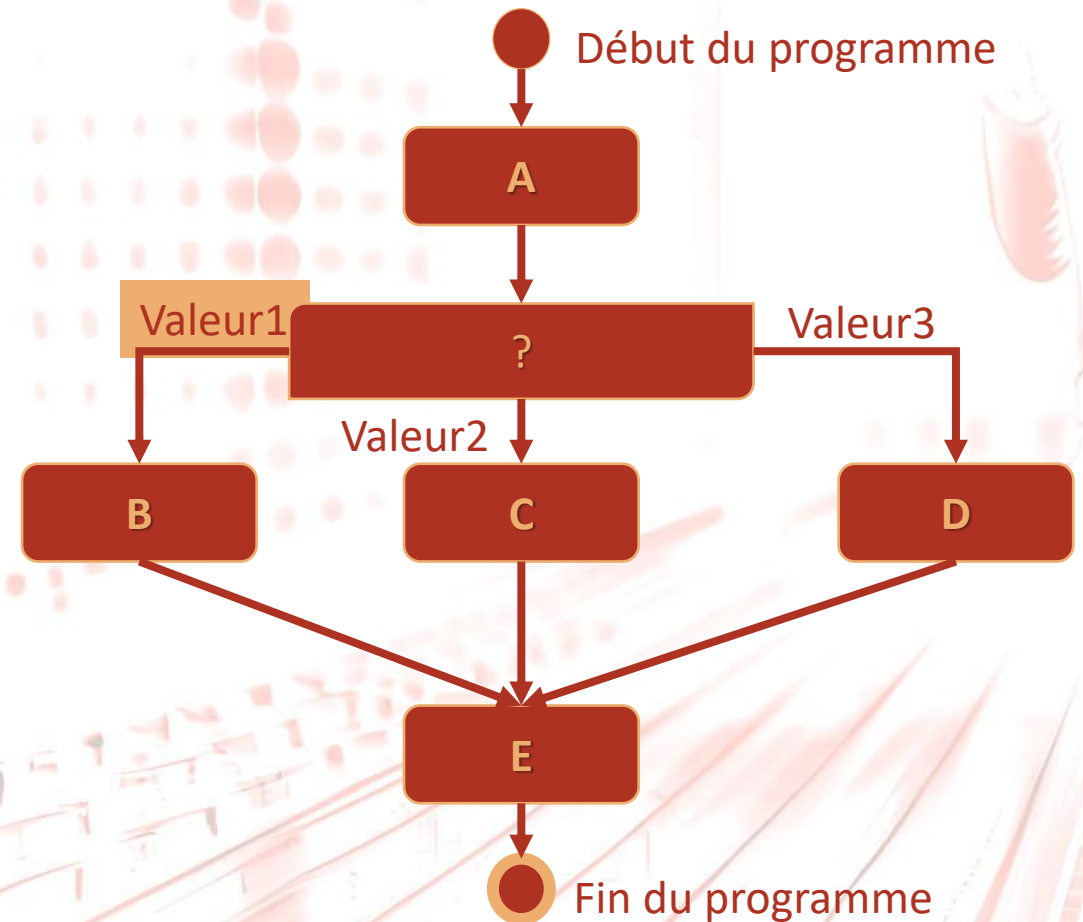


CONDITIONS: SWITCH CASE

Programmation - Les structures

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        default: nb=3;
    }
    return 0;
}
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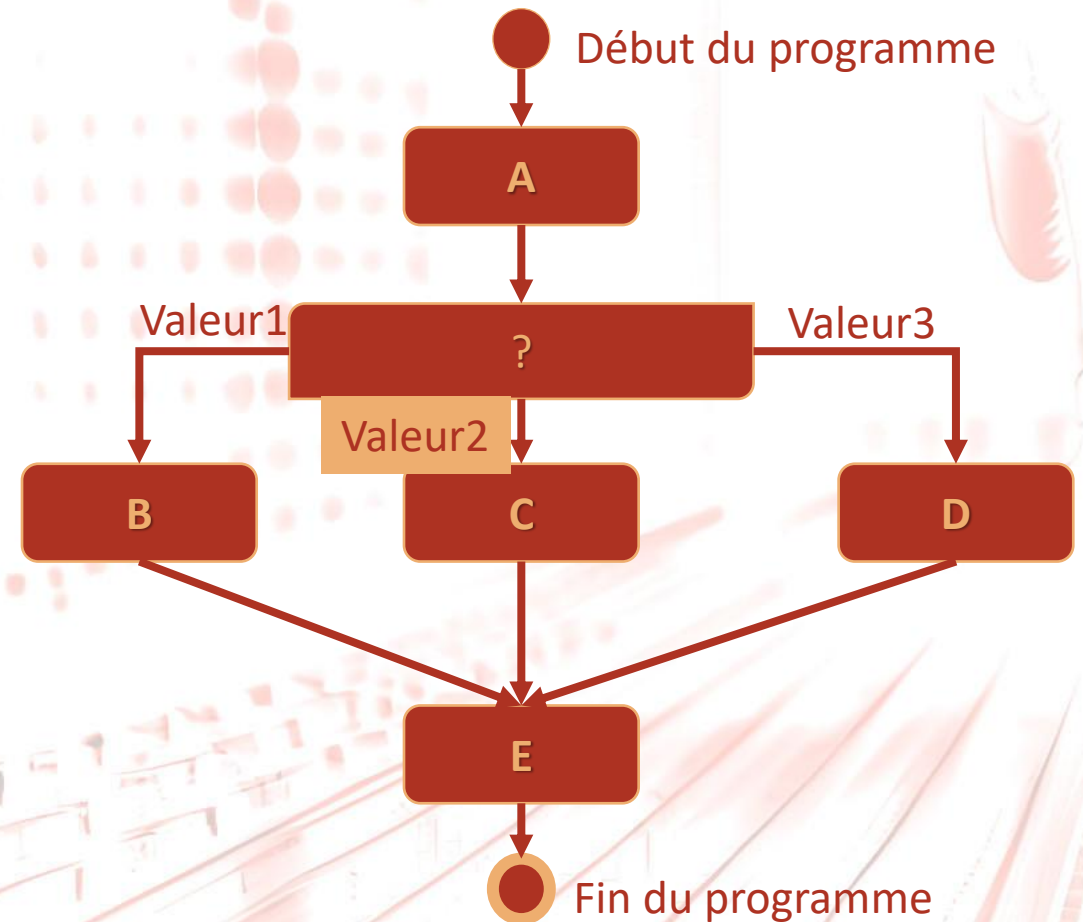


CONDITIONS: SWITCH CASE

Programmation - Les structures

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    switch(nb)
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        default: nb=3;
    }
    return 0;
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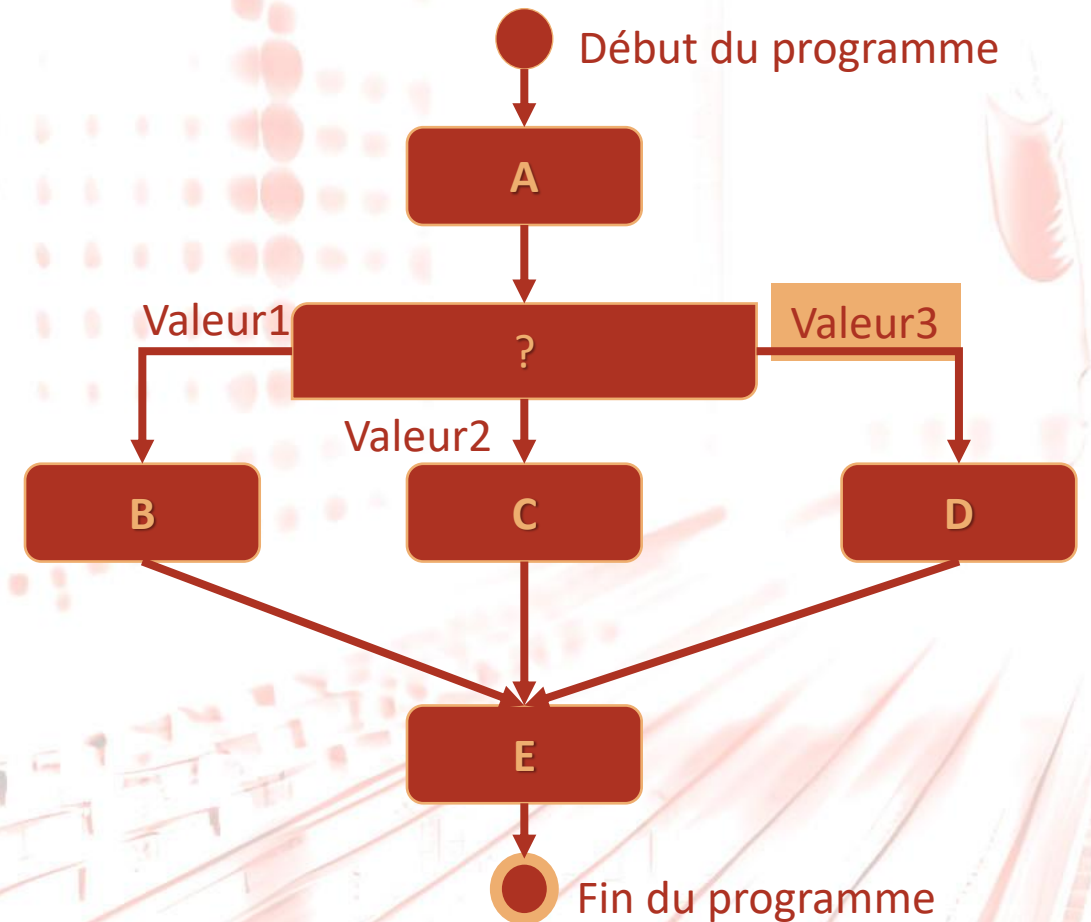


CONDITIONS: SWITCH CASE

Programmation - Les structures

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    }
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```

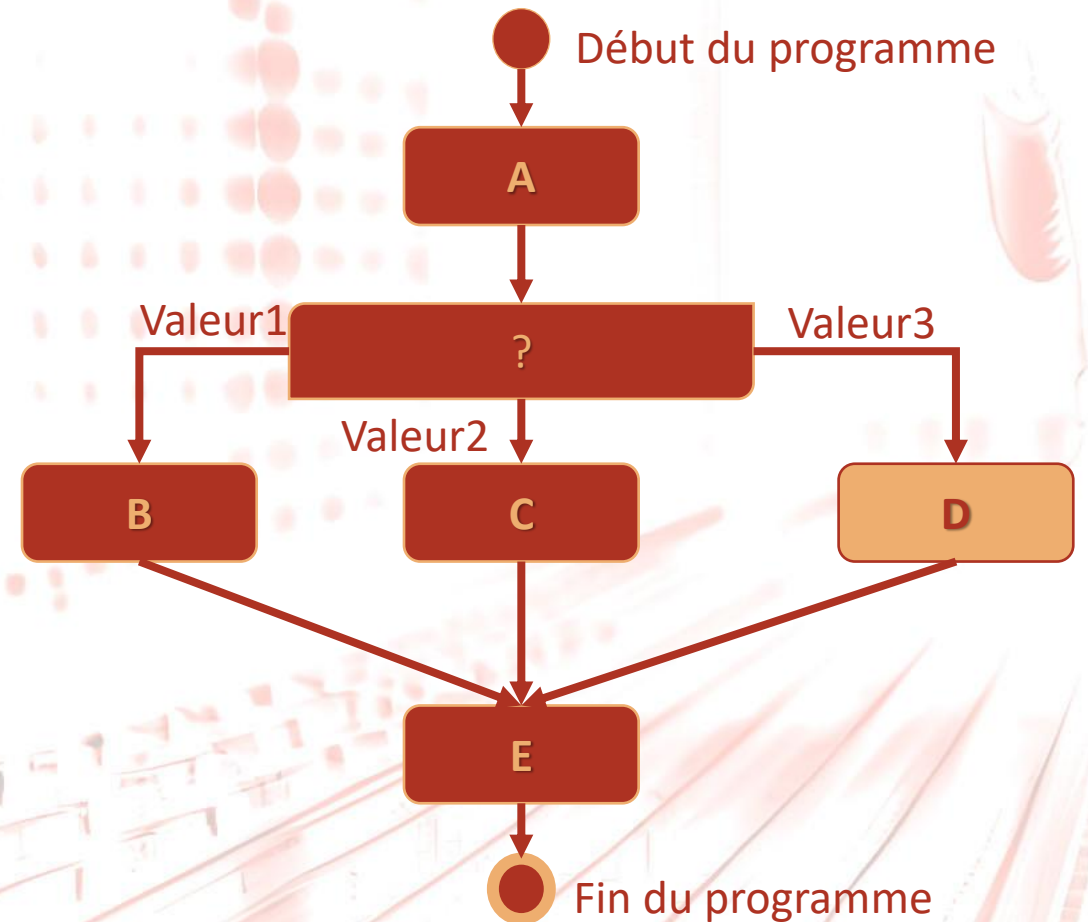


CONDITIONS: SWITCH CASE

Programmation - Les structures

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using namespace std;

int main() {
    int nb = 4;
    switch(nb)
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        default: nb=3;
    }
    return 0;
}
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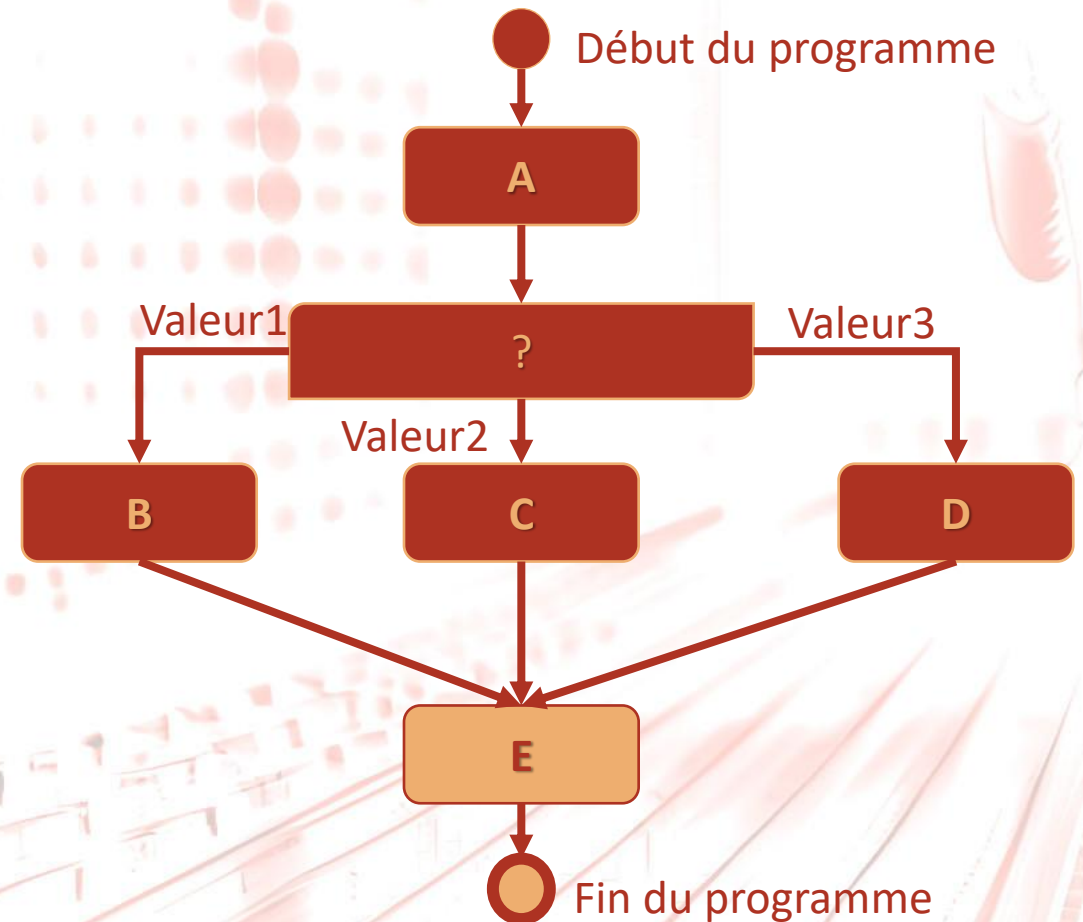


CONDITIONS: SWITCH CASE

Programmation - Les structures

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        default: nb=3;
    }
    return 0;
}
```



ITÉRATIONS

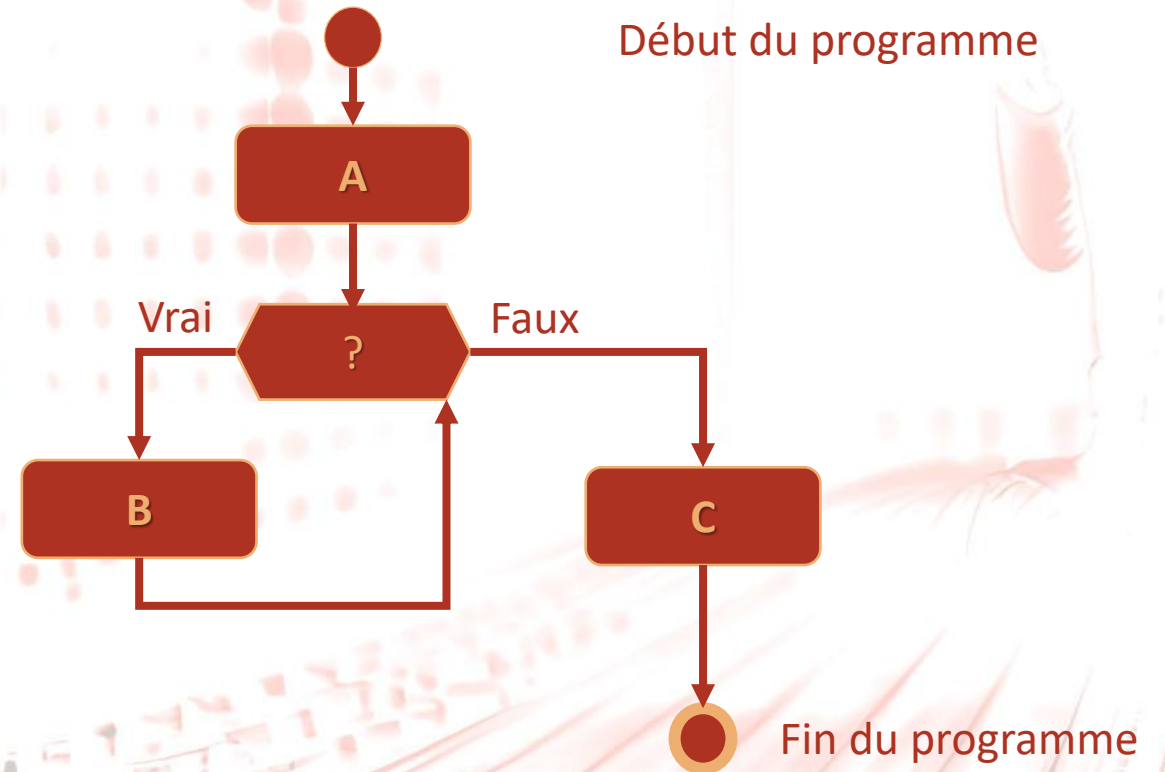
Programmation - Les structures



ITÉRATIONS

Programmation - Les structures

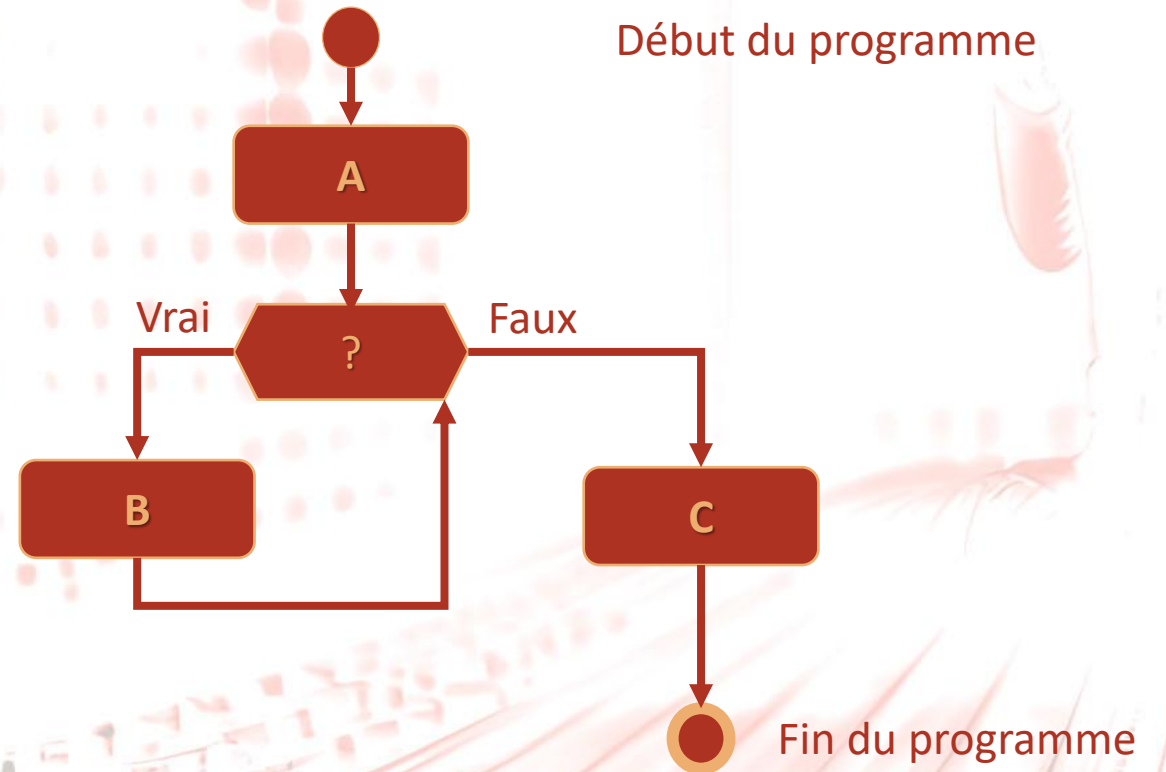
- Les itérations sont composées de séquences qui sont exécutées plusieurs fois.
- On les appelle plus simplement boucles.
- Le nombre d'itérations dépend d'une ou de plusieurs conditions.
- En C++, il en existe 3 types:
 - while
 - do while
 - for



ITÉRATIONS: WHILE

Programmation - Les structures

- ✎ Dans le cas du while, la condition est testée en premier.
- ✎ Si la condition est fausse au départ, la séquence n'est pas du tout exécutée.

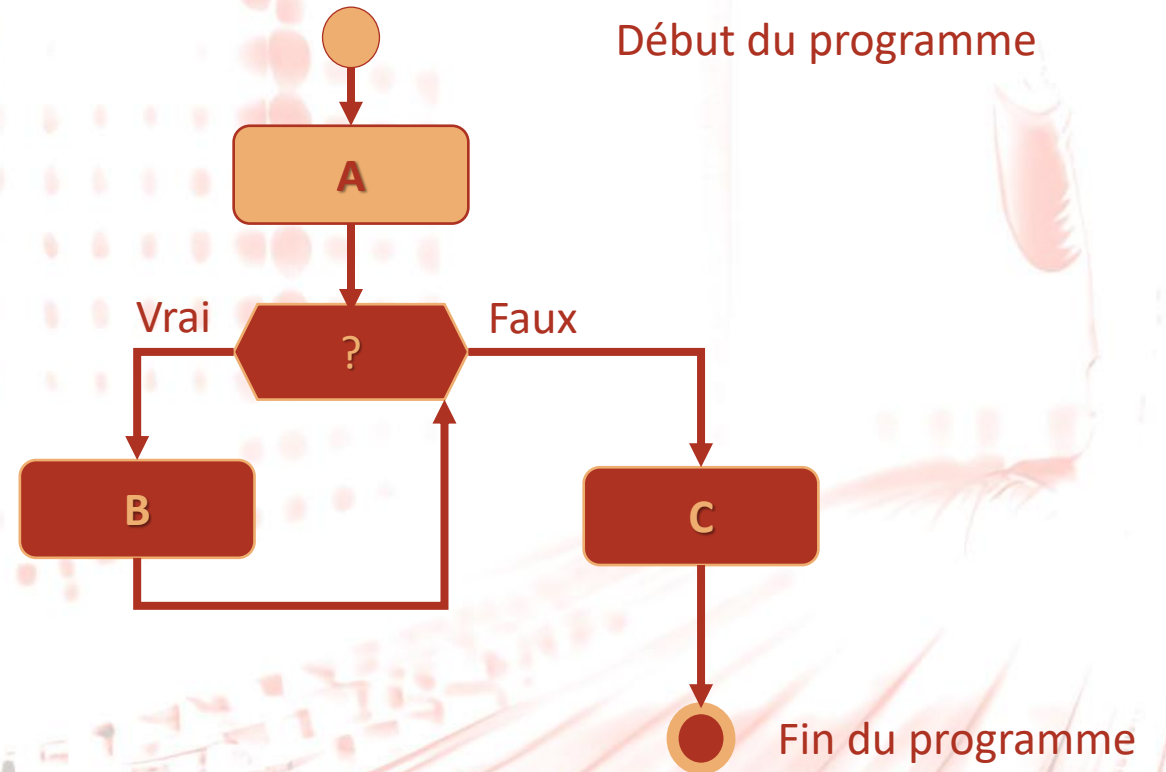


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb>0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```

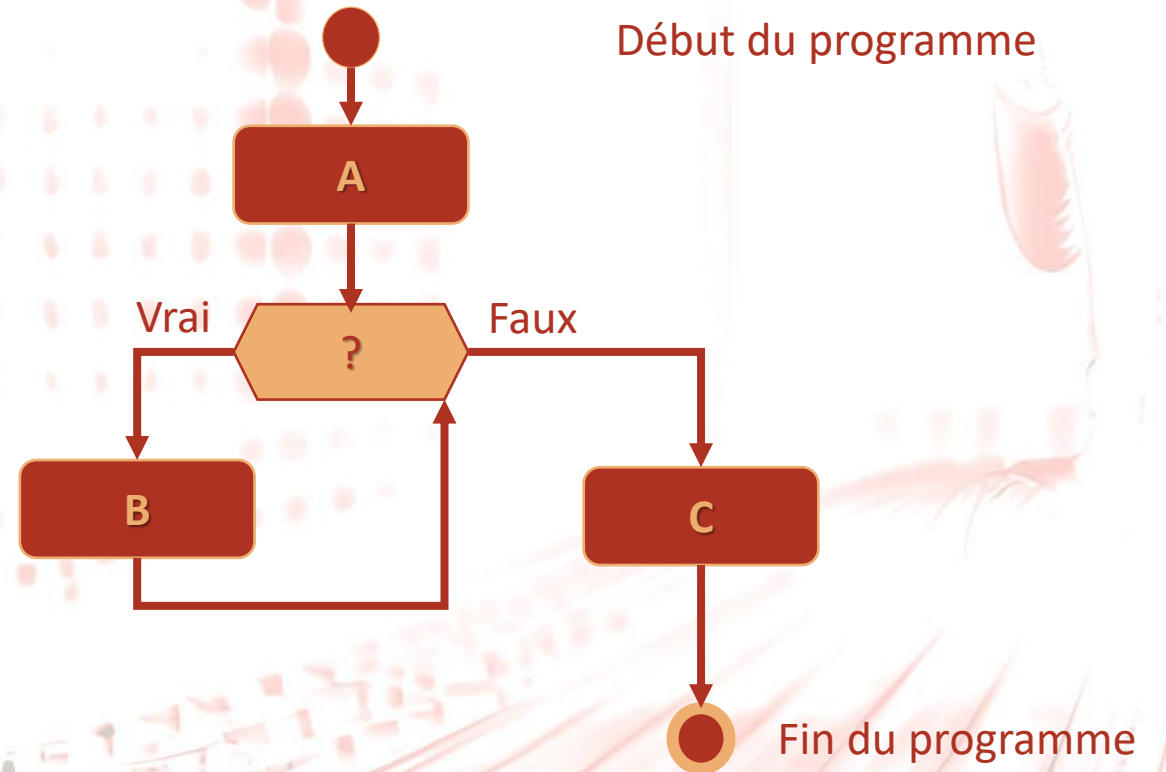


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb>0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```

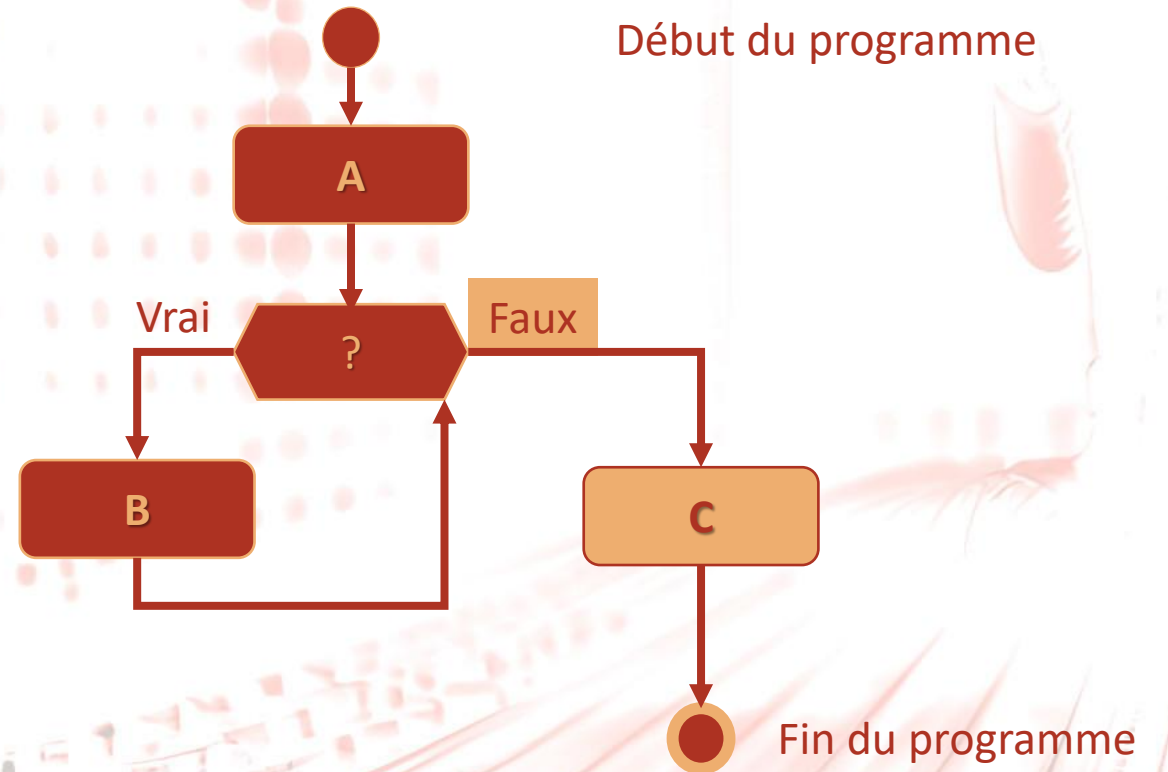


ITÉRATIONS: WHILE

Programmation - Les structures

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using namespace std;

int main(){
    int nb = 0;
    while(nb>0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```

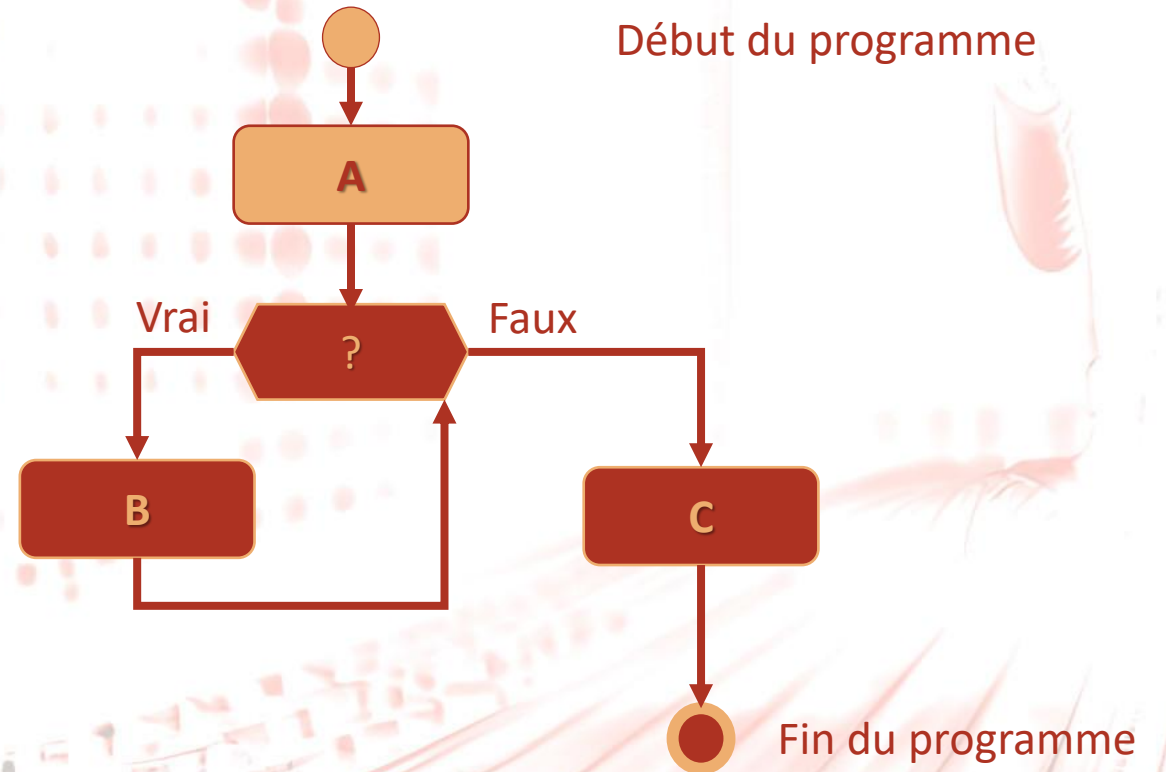


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb<=0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```

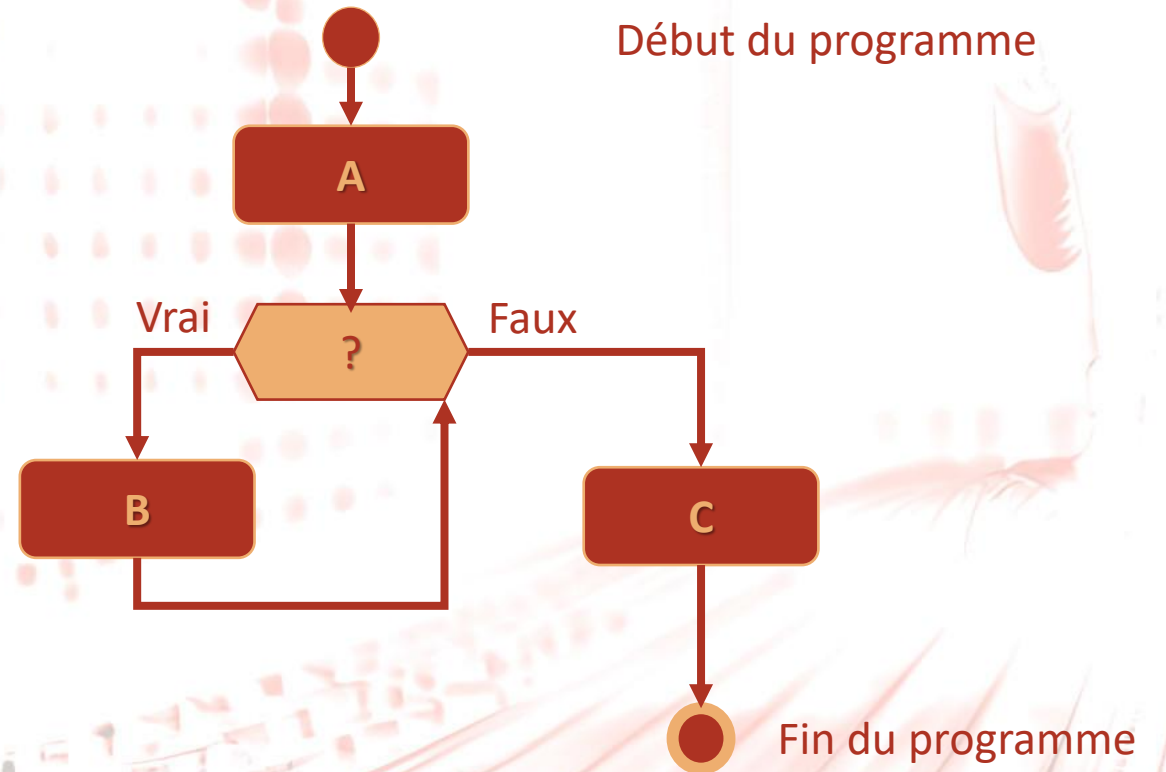


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb<=0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```

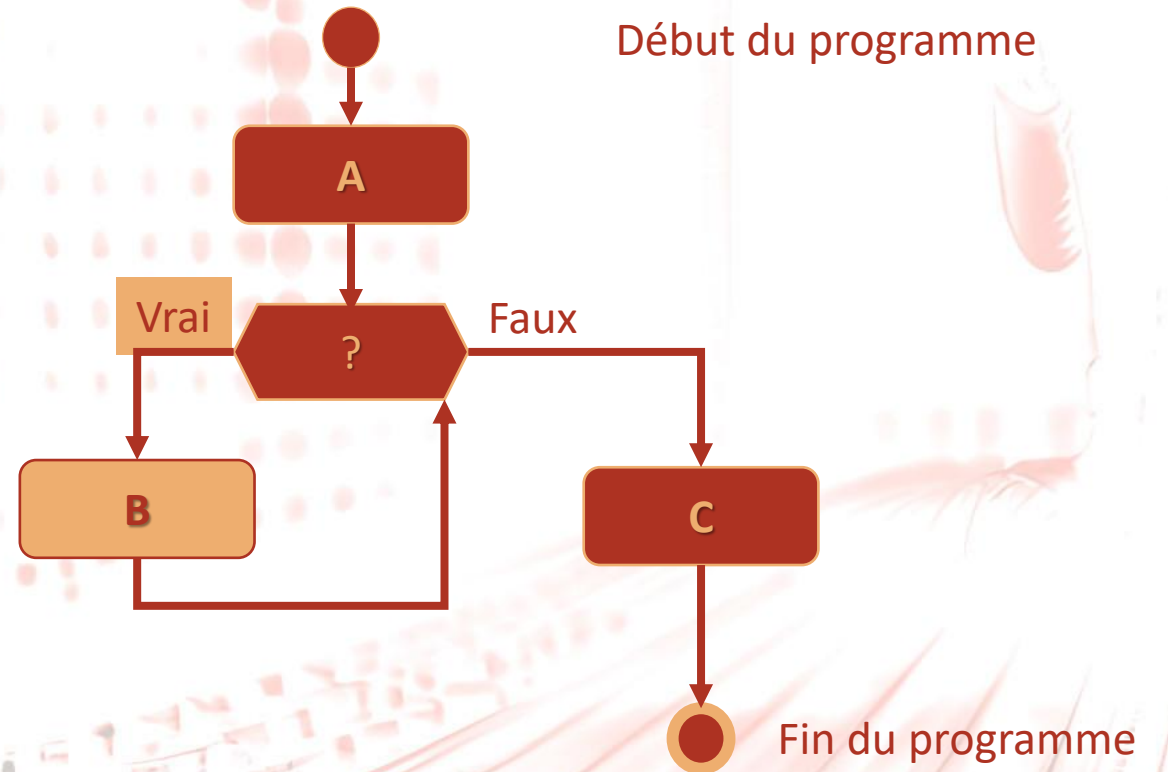


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb<=0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
        // l'utilisateur saisit 0
    }
    return 0;
}
```

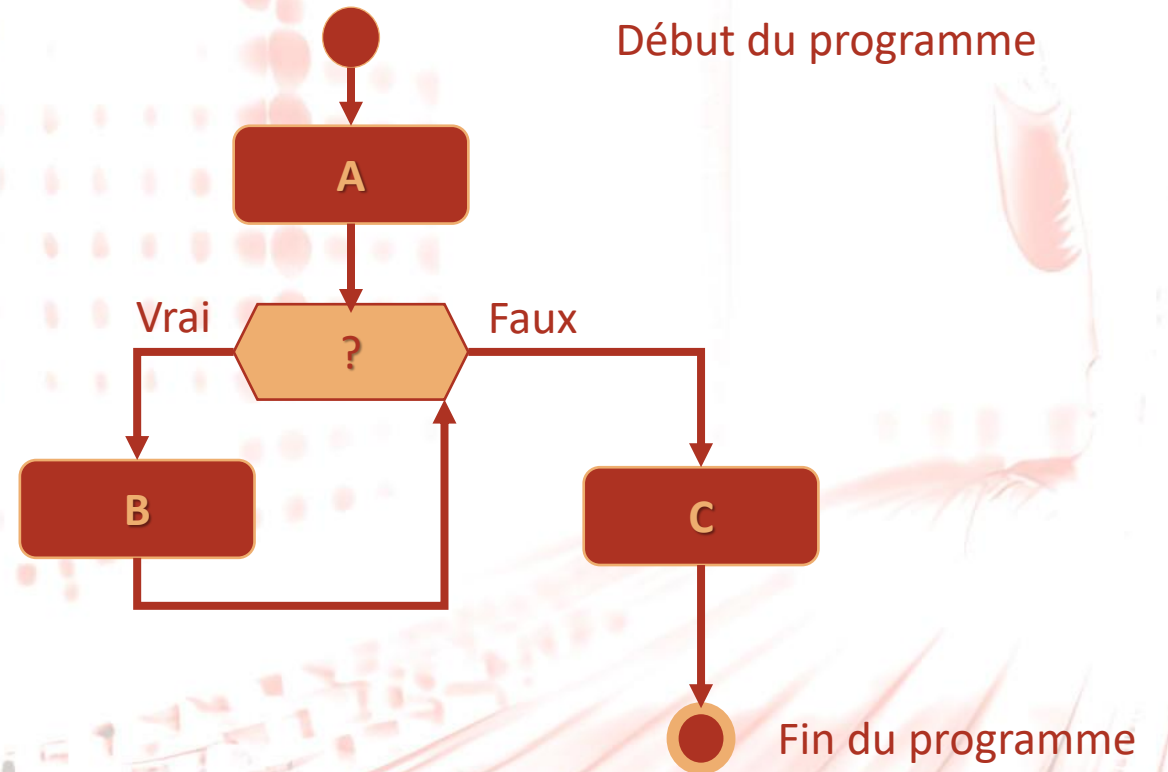


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb<=0) //nb vaut 0
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```

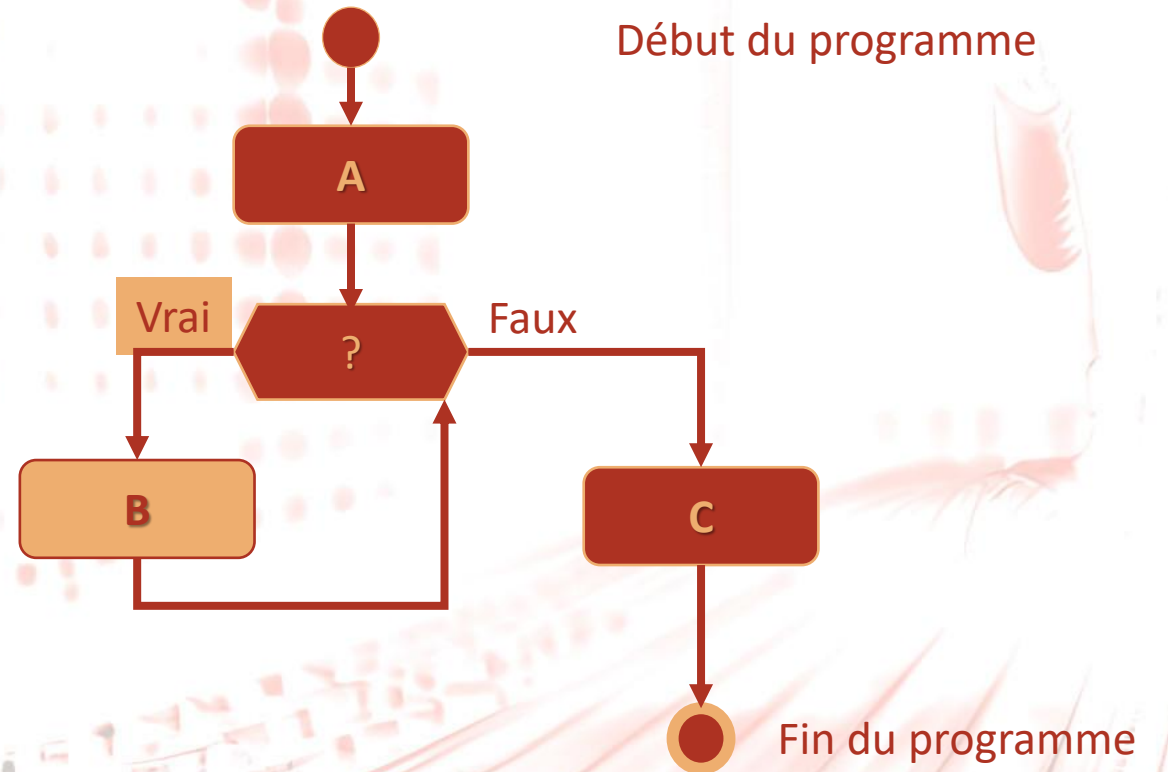


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb<=0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
        // l'utilisateur saisit 1
    }
    return 0;
}
```

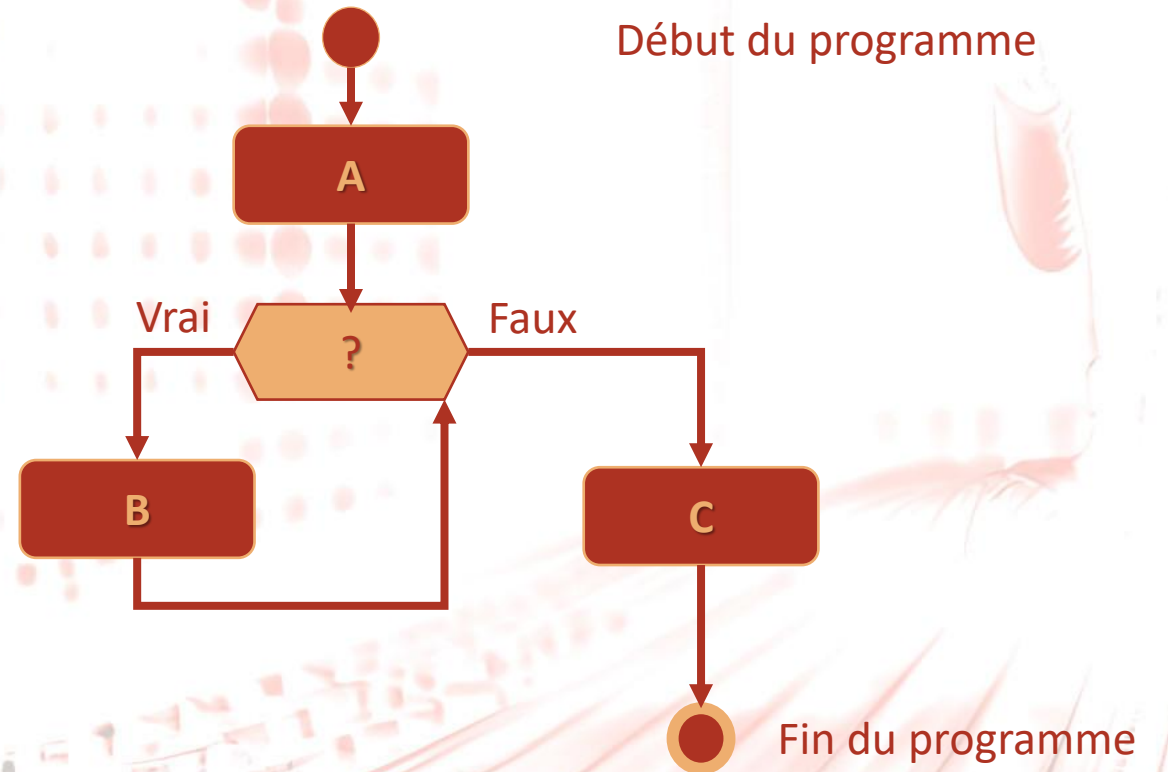


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb = 0;
    while(nb<=0) //nb vaut 1
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```

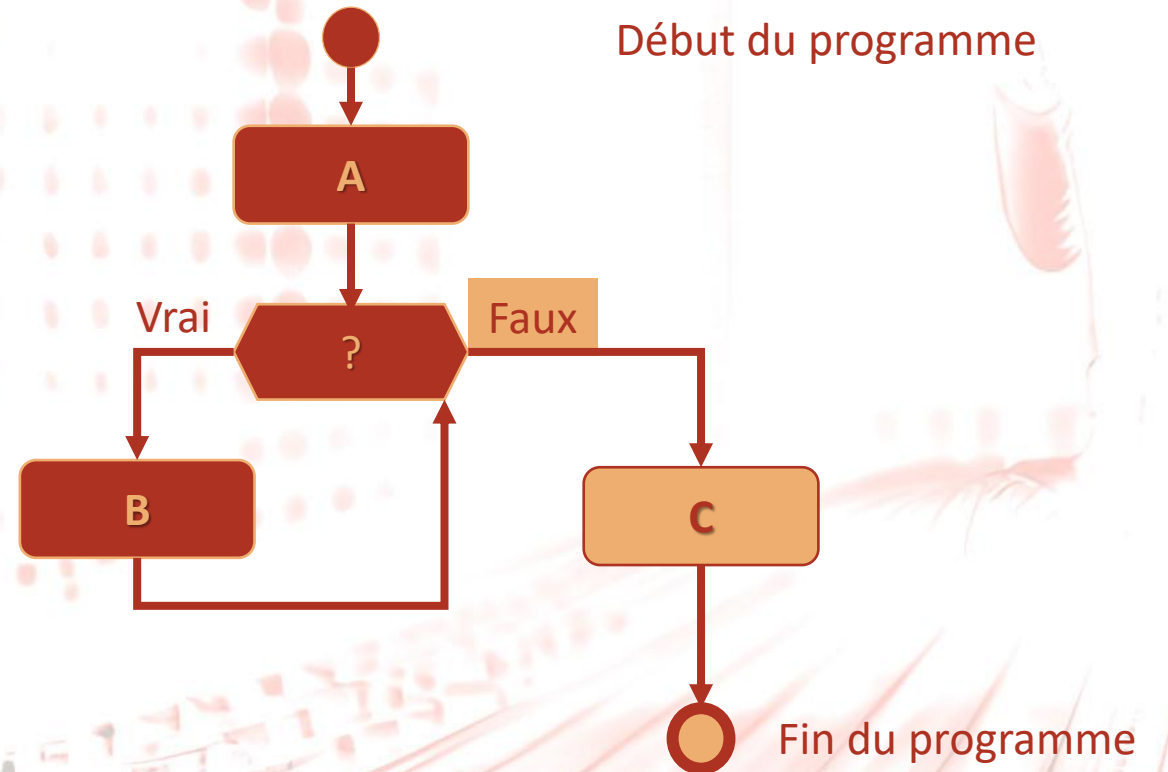


ITÉRATIONS: WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

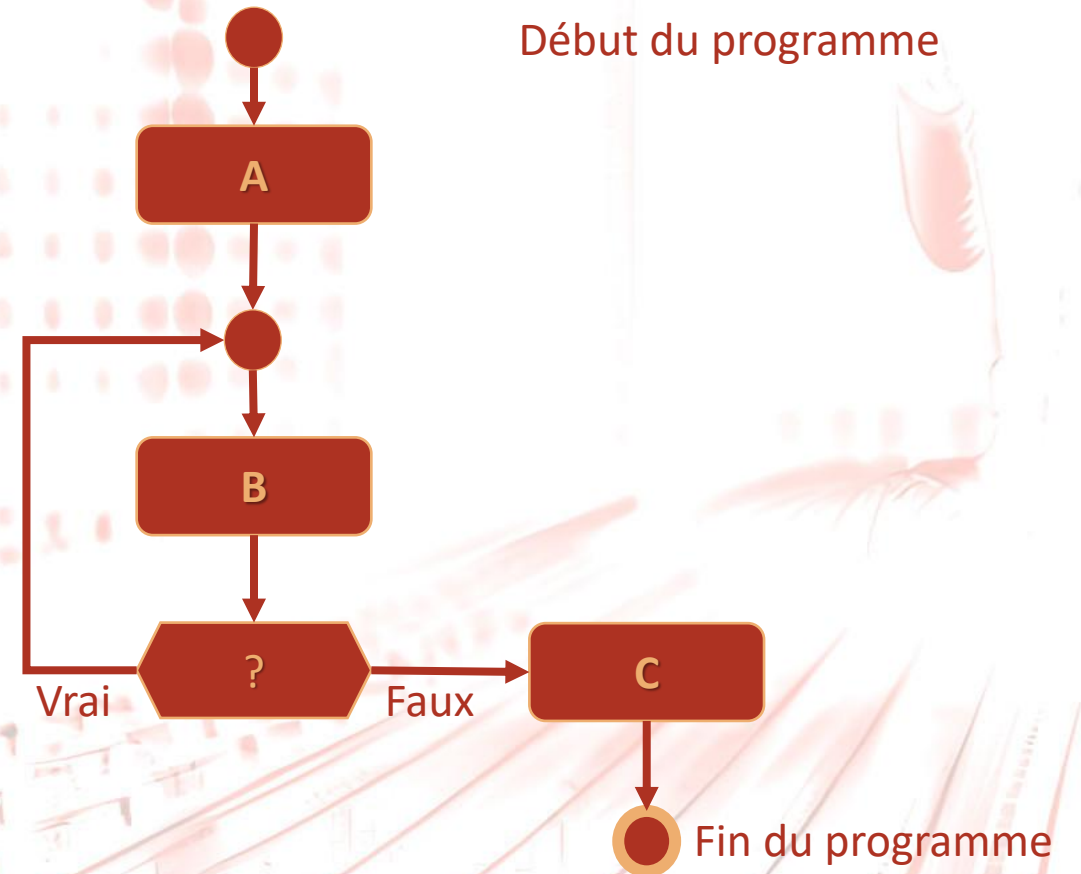
int main(){
    int nb = 0;
    while(nb<0)
    {
        cout<<"nb?"<<endl;
        cin>>nb;
    }
    return 0;
}
```



ITÉRATIONS: DO WHILE

Programmation - Les structures

- Le do while permet d'exécuter au moins une fois la séquence qu'il contient.
- Si la condition est fausse au départ, la séquence est exécutée malgré tout une fois au minimum.

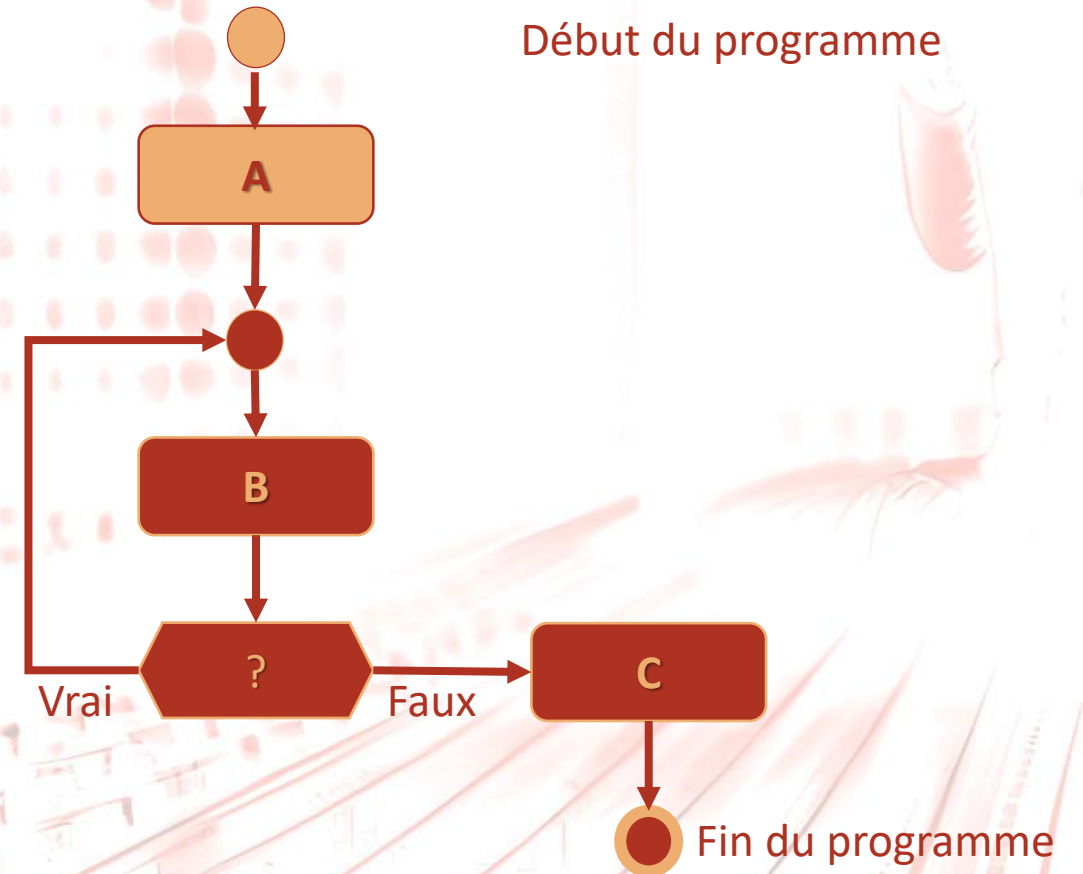


ITÉRATIONS: DO WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    char nb = 'n';
    do{
        cout<<"o/n?"<<endl;
        cin>>nb;
    } while(nb=='o');
    return 0;
}
```

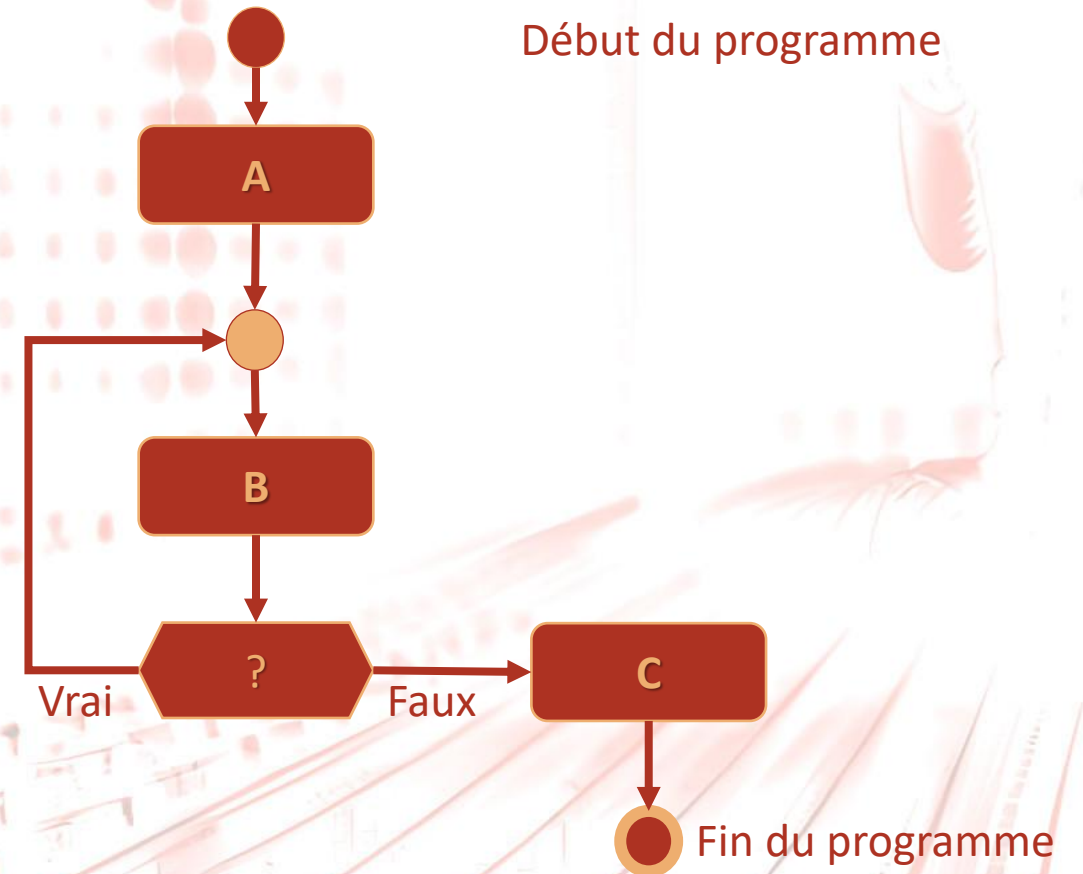


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Programmation - Les structures

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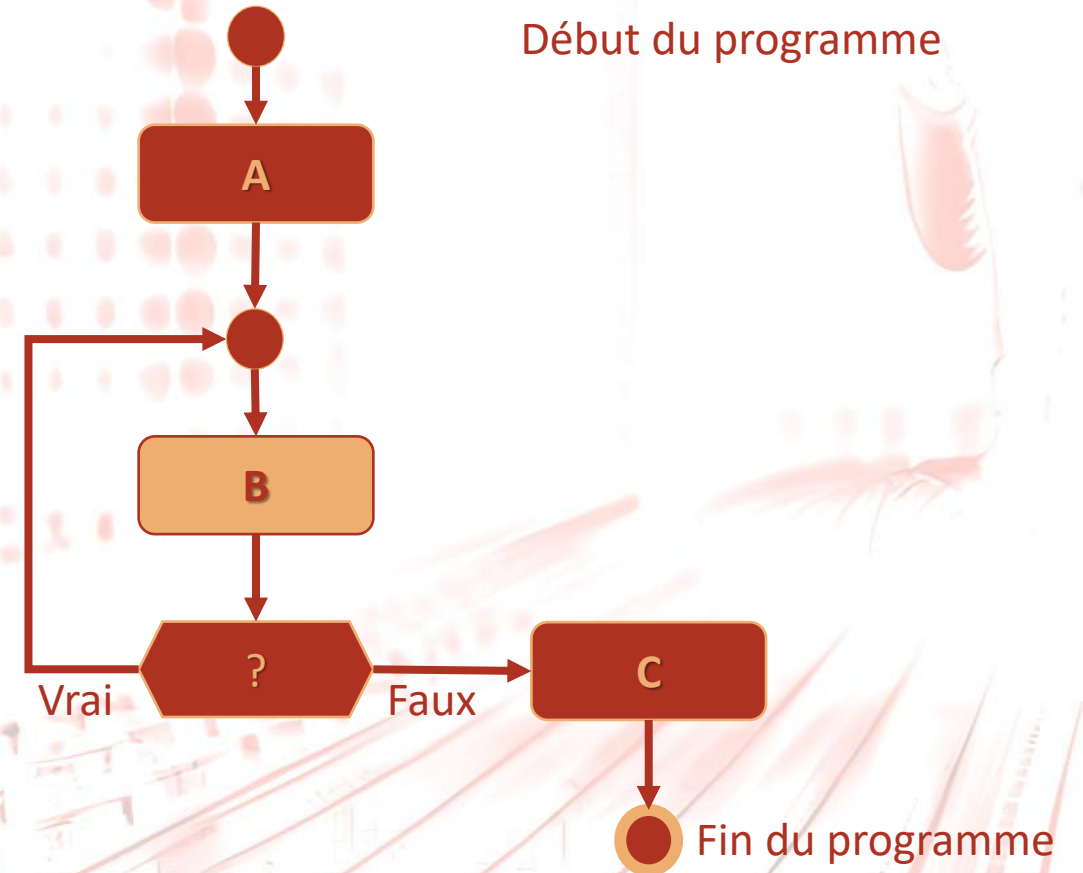


ITÉRATIONS: DO WHILE

Programmation - Les structures

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using namespace std;

int main(){
    char nb = 'n';
    do{
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        cin>>nb;
        //l'utilisateur saisit 'o'
    } while(nb=='o');
    return 0;
}
```

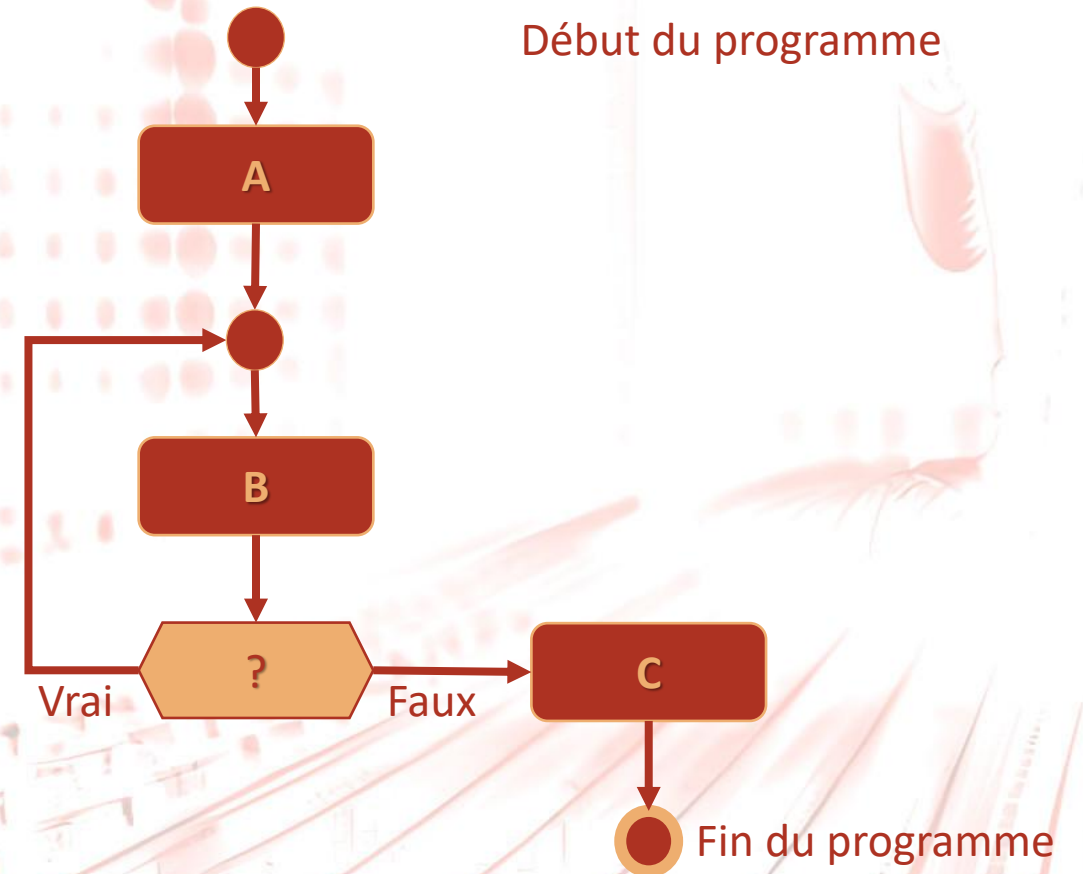


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```

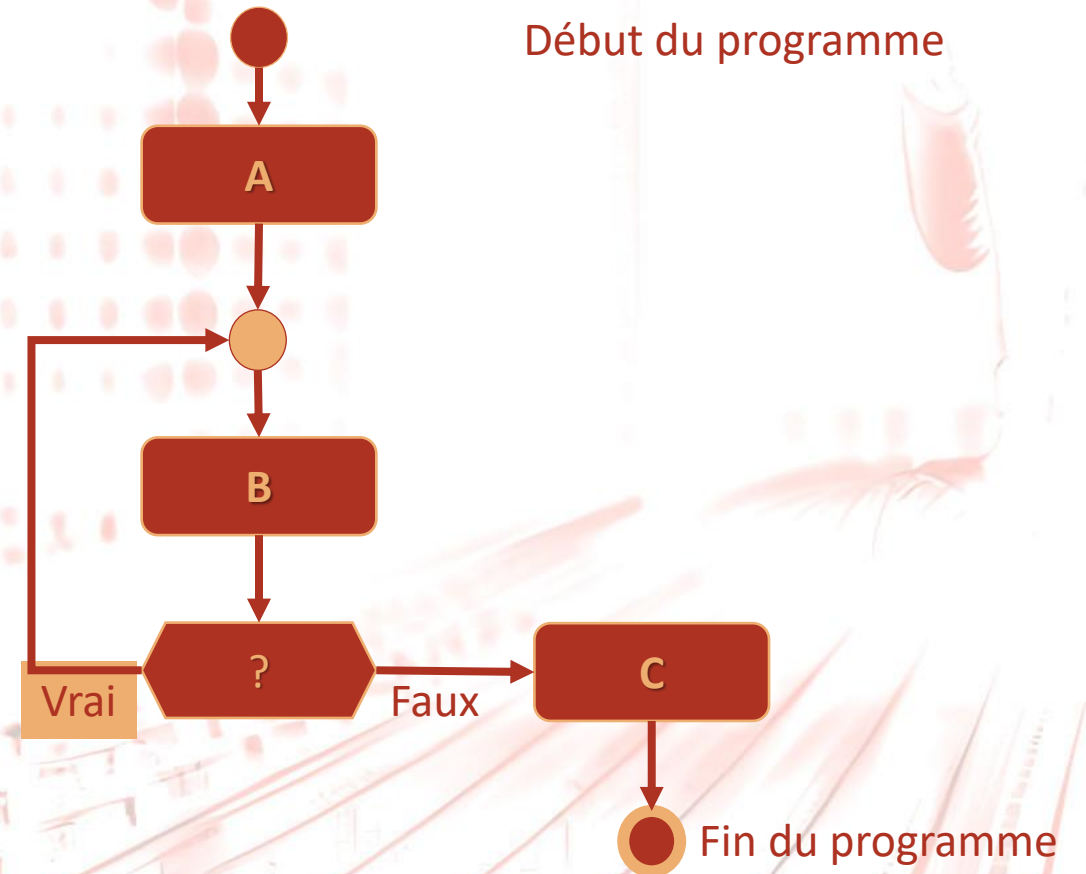


ITÉRATIONS: DO WHILE

Programmation - Les structures

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using namespace std;

int main(){
    char nb = 'n';
    do{
        cout<<"o/n?"<<endl;
        cin>>nb;
    } while(nb=='o');
    return 0;
}
```

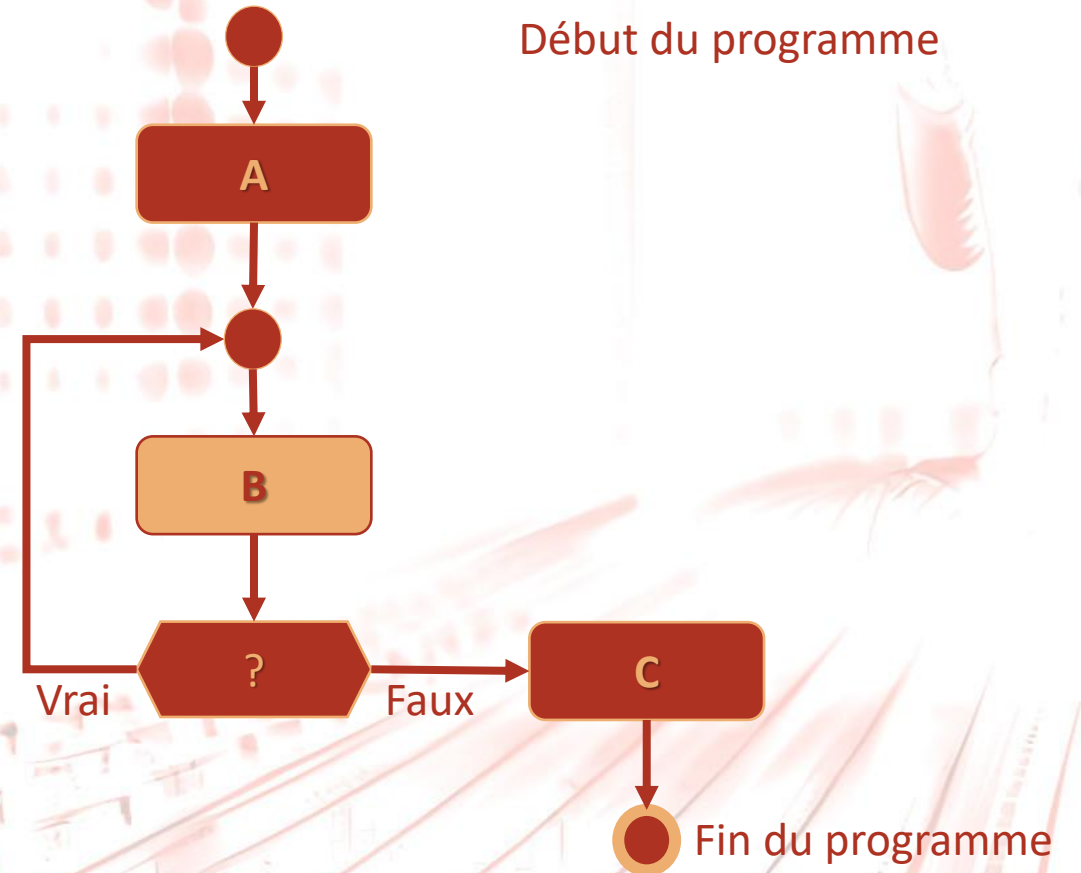


ITÉRATIONS: DO WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    char nb = 'n';
    do{
        cout<<"o/n?"<<endl;
        cin>>nb;
        //l'utilisateur saisit 'n'
    } while(nb=='o');
    return 0;
}
```

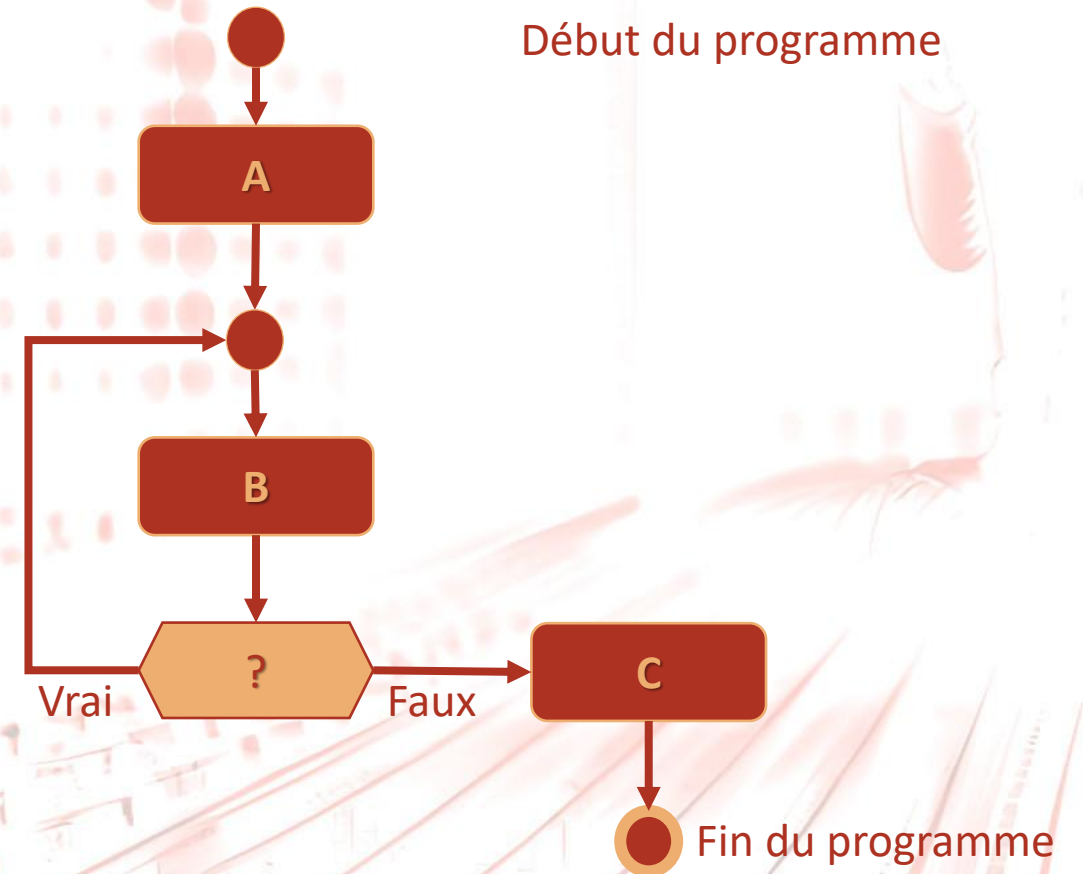


ITÉRATIONS: DO WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    char nb = 'n';
    do{
        cout<<"o/n?"<<endl;
        cin>>nb;
    } while(nb=='o');
    return 0;
}
```

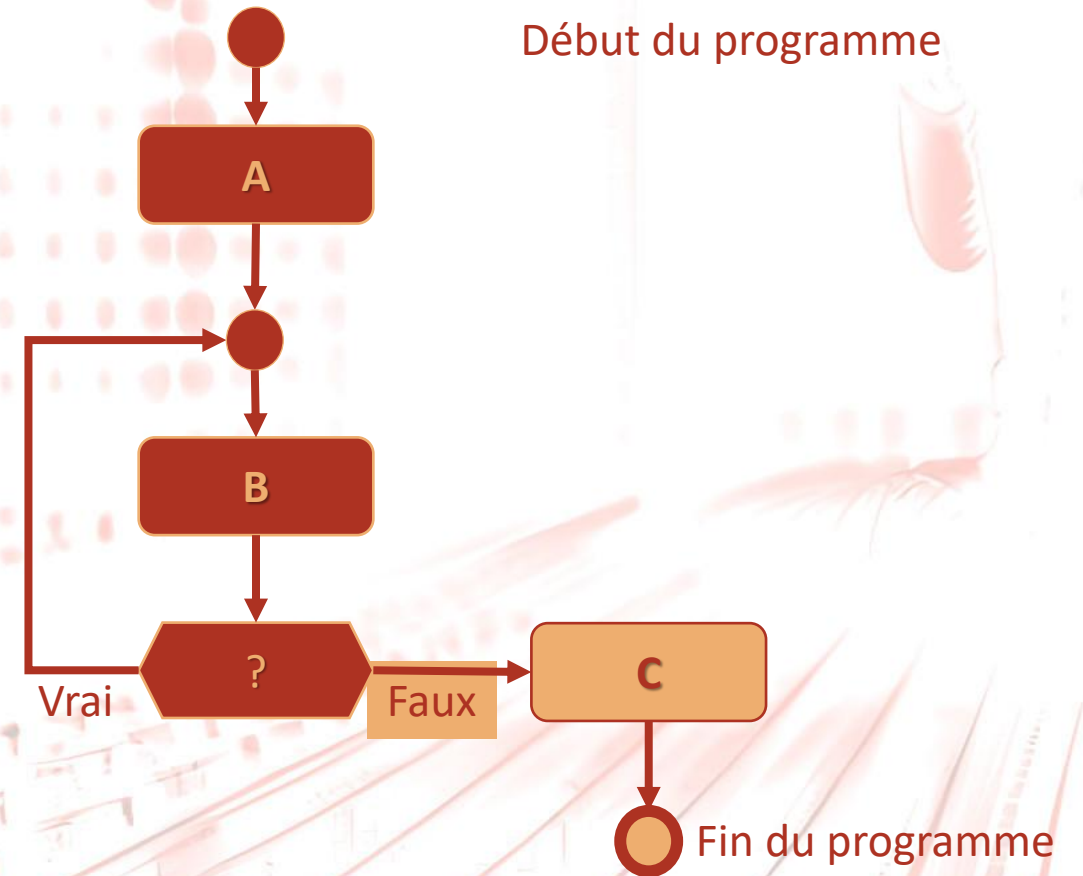


ITÉRATIONS: DO WHILE

Programmation - Les structures

```
#include <iostream>
using namespace std;

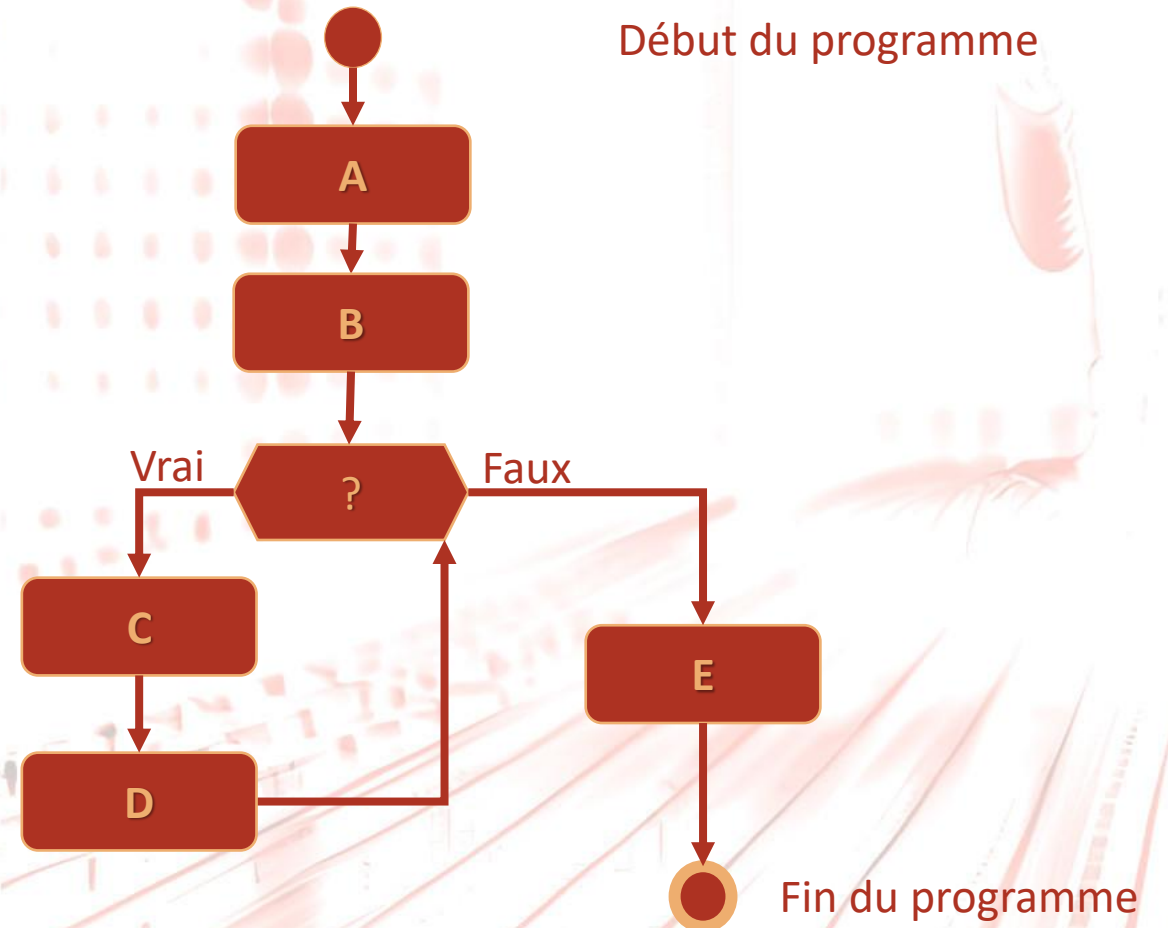
int main(){
    char nb = 'n';
    do{
        cout<<"o/n?"<<endl;
        cin>>nb;
    } while(nb=='o');
    return 0;
}
```



ITÉRATIONS: FOR

Programmation - Les structures

- Le for permet de boucler un nombre défini de fois.
- C'est une boucle très pratique dès lors que l'on veut automatiser un traitement de données rangées séquentiellement.



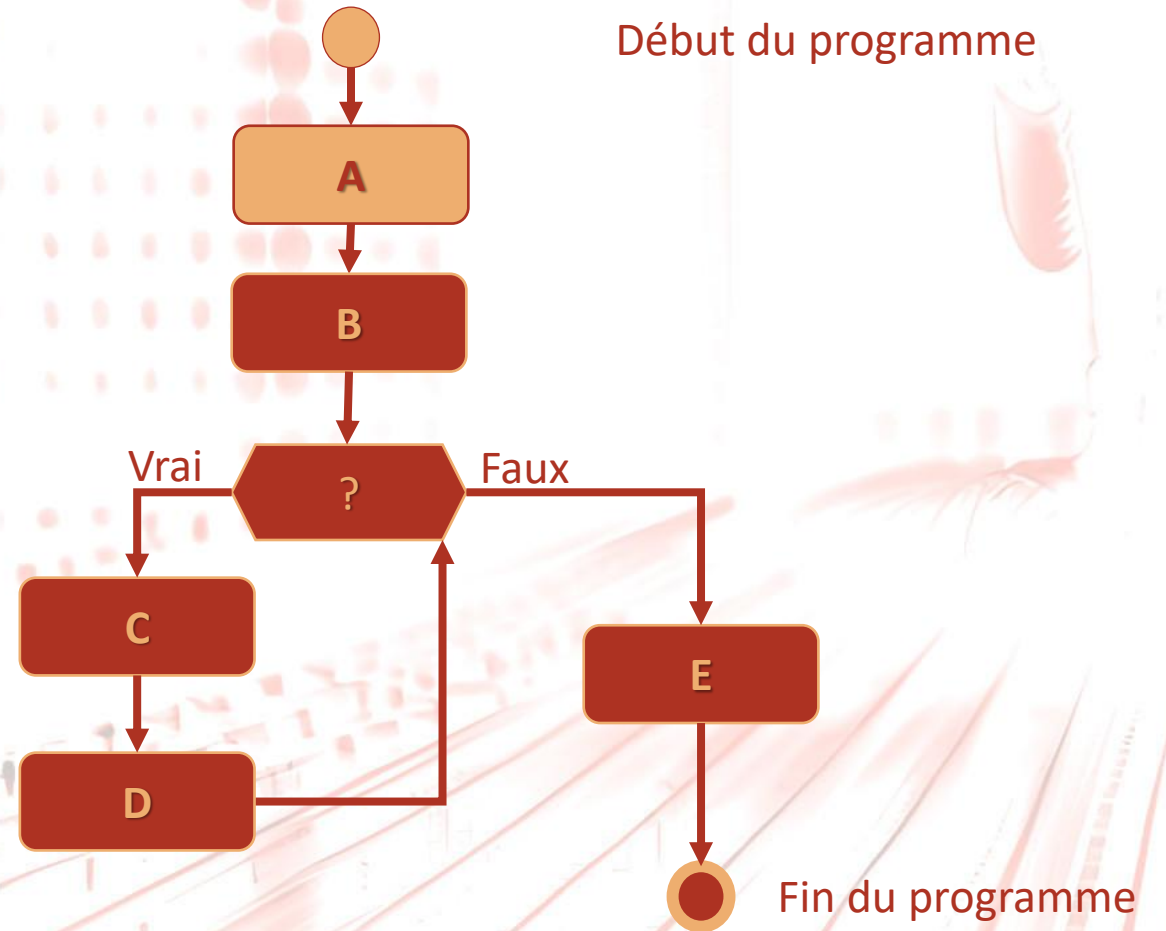
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}
```

nb[0] vaut 0
nb[1] vaut 0
nb[2] vaut 0
i n'existe pas encore



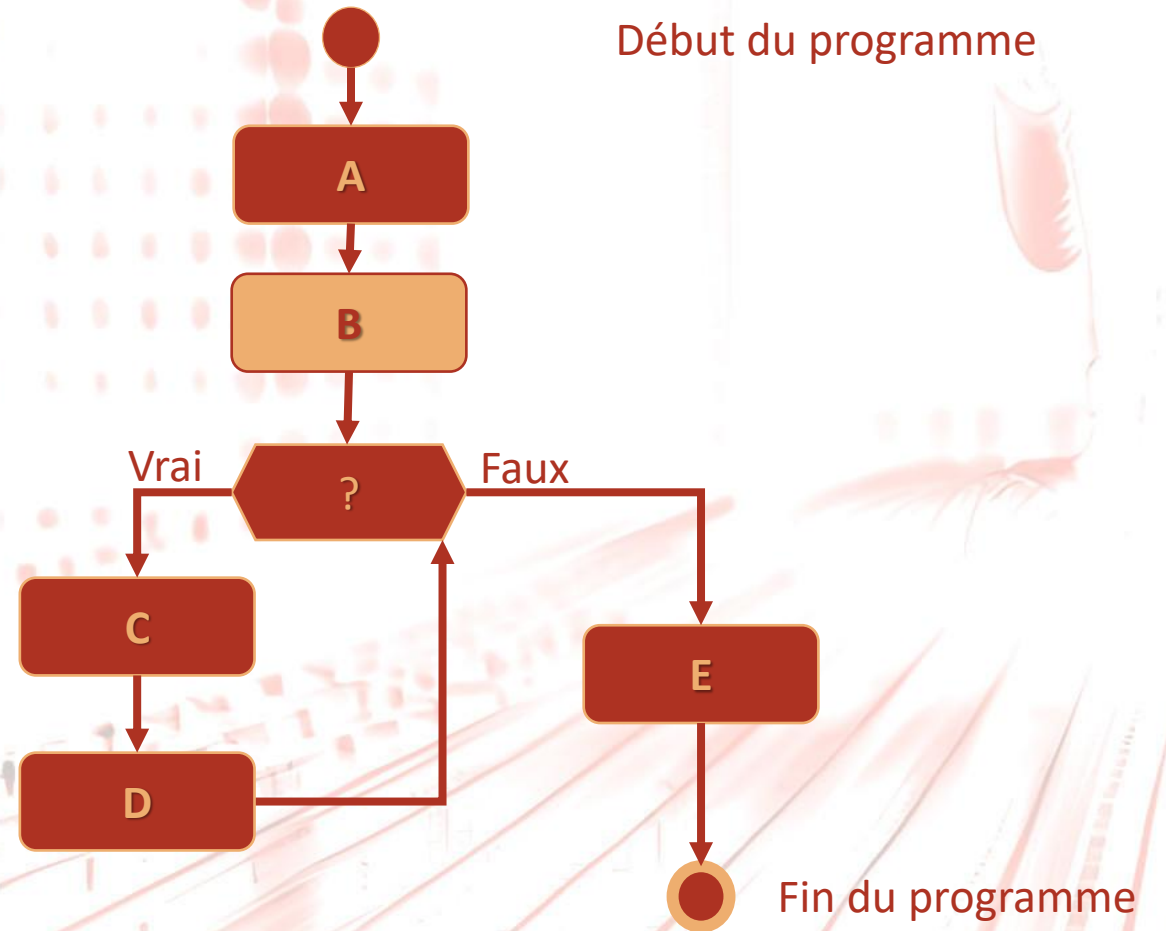
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 0
nb[1] vaut 0
nb[2] vaut 0
i vaut 0
```



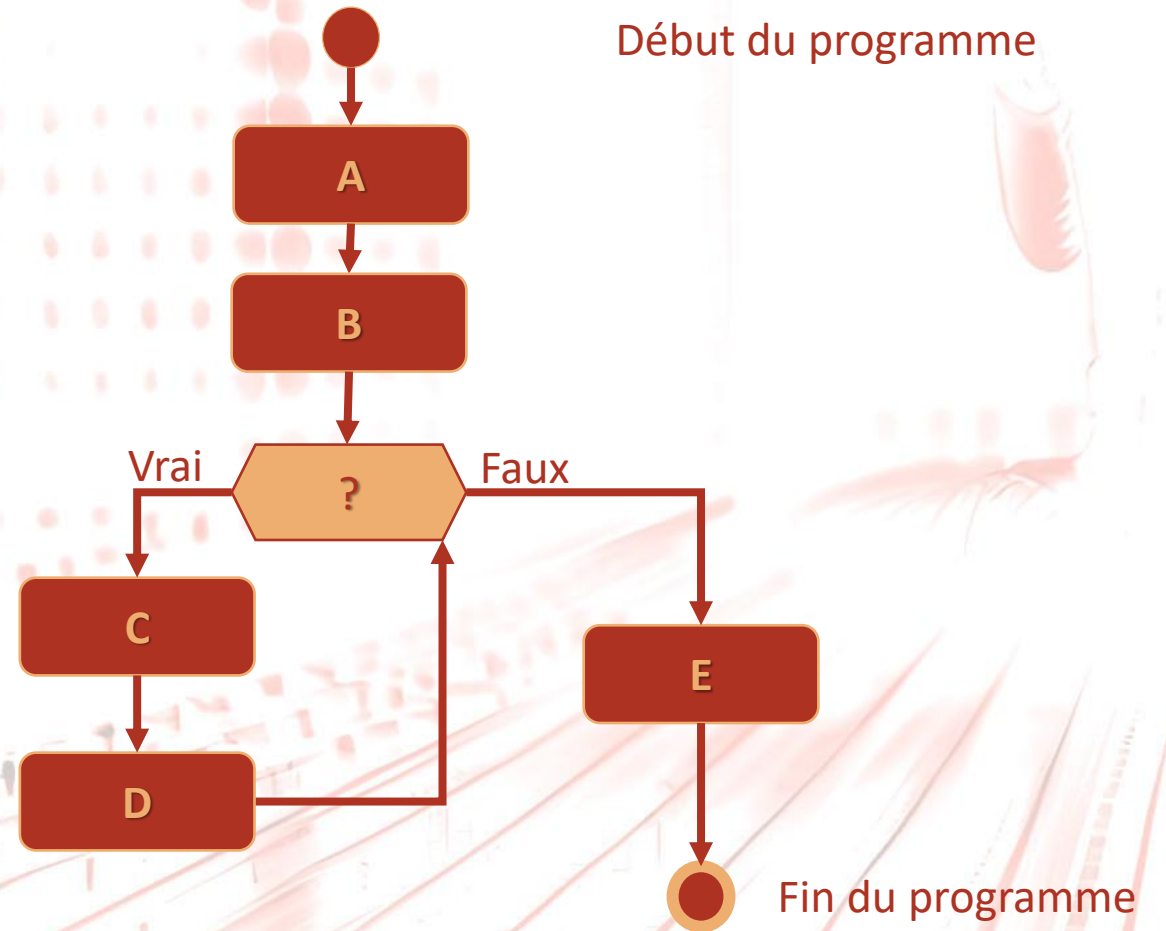
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0; i<3; i++){
        nb[i]=i+1;
    }
    return 0;
}
```

```
nb[0] vaut 0
nb[1] vaut 0
nb[2] vaut 0
i vaut 0
```



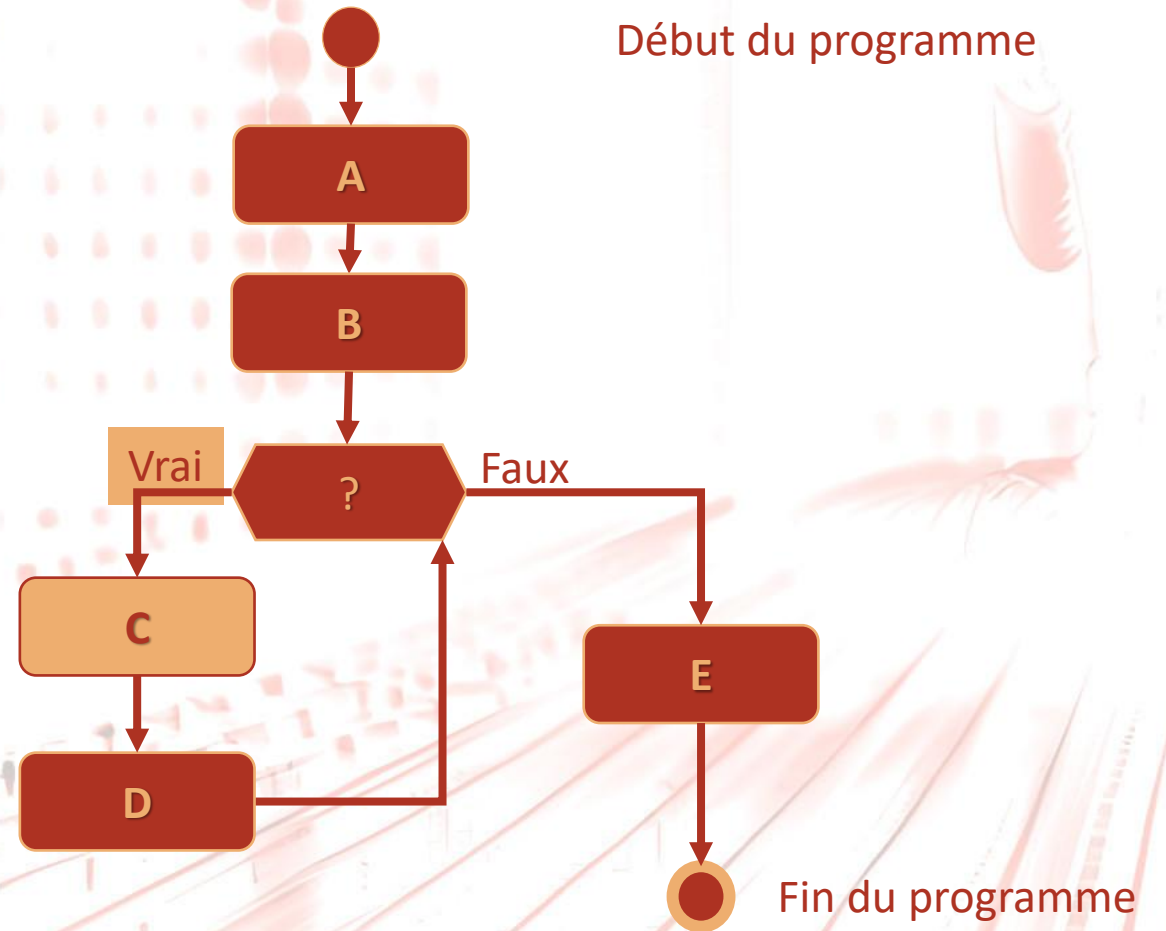
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 0
nb[2] vaut 0
i vaut 0
```



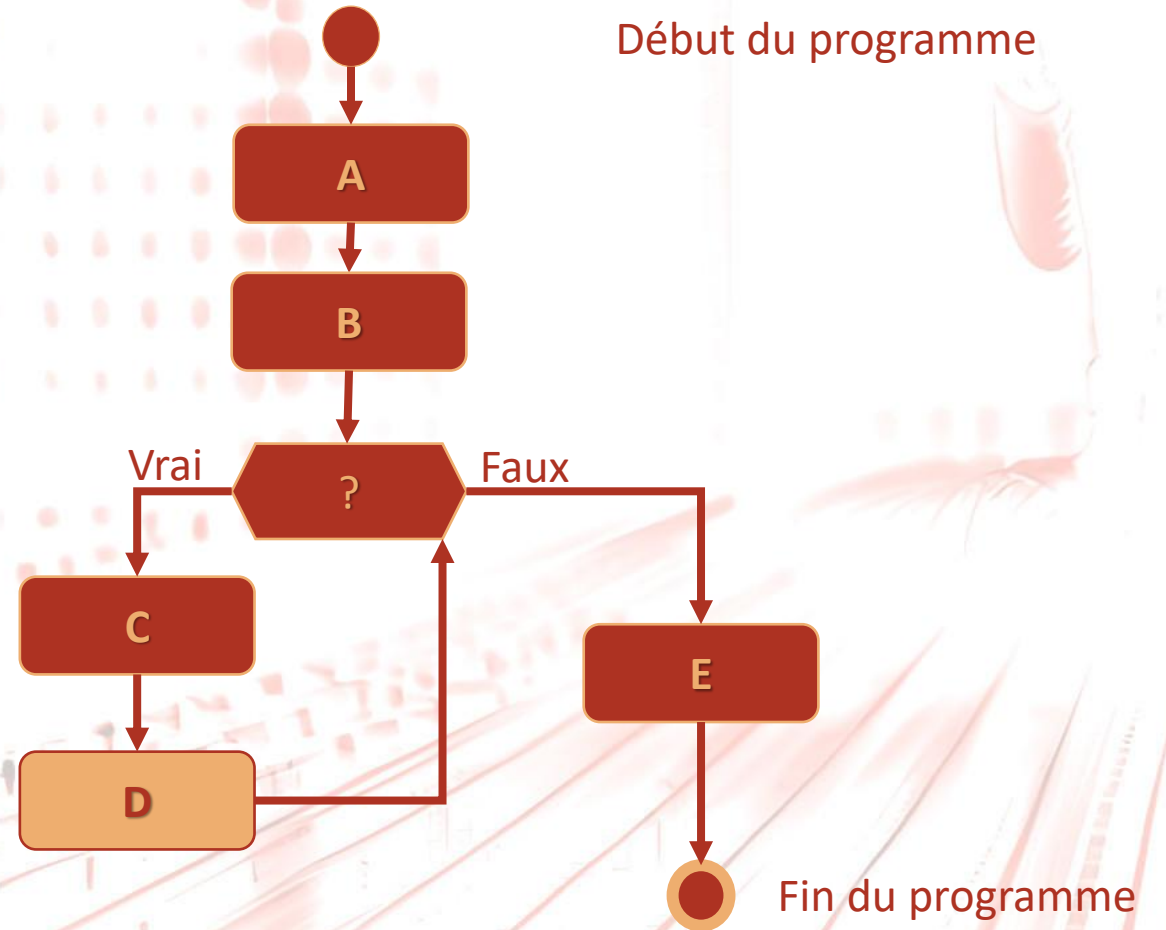
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 0
nb[2] vaut 0
i vaut 1
```



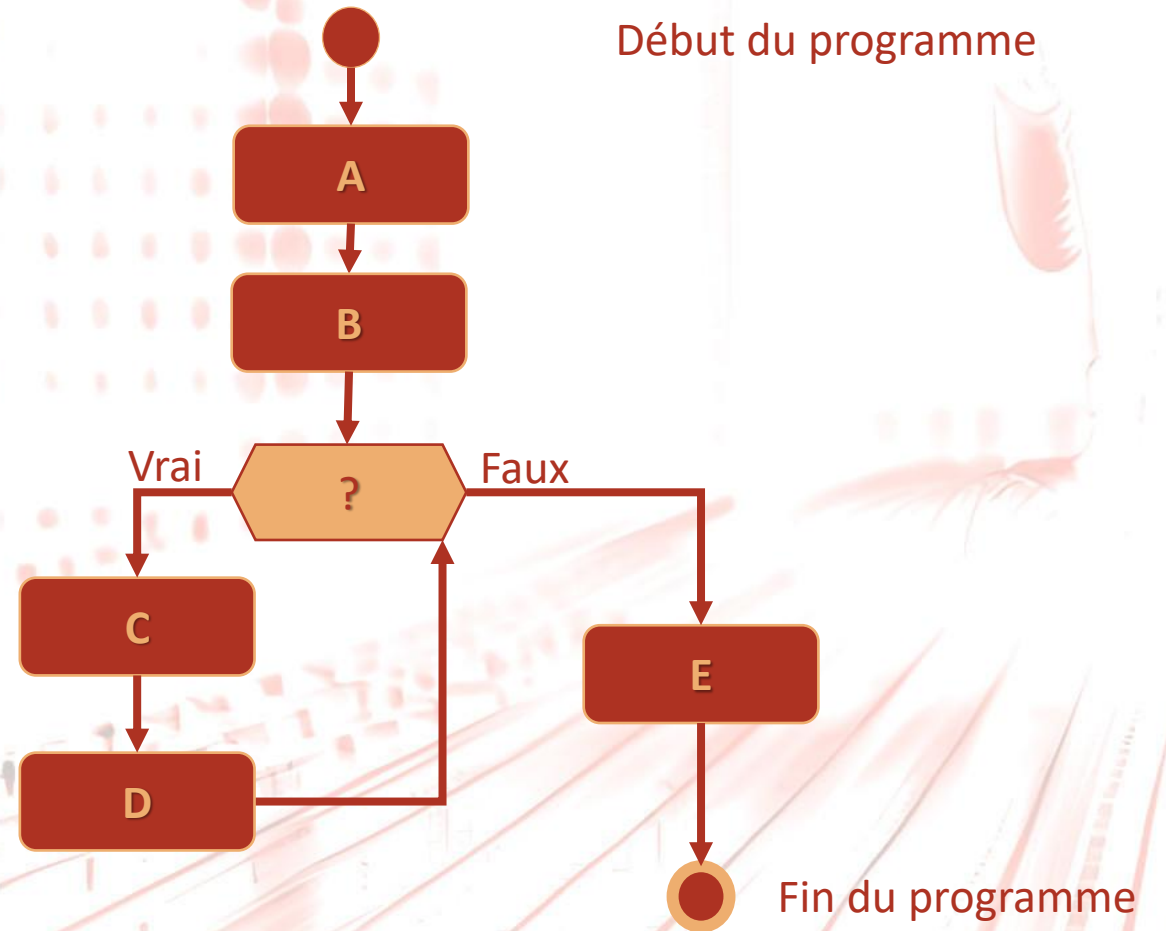
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0; i<3; i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 0
nb[2] vaut 0
i vaut 1
```



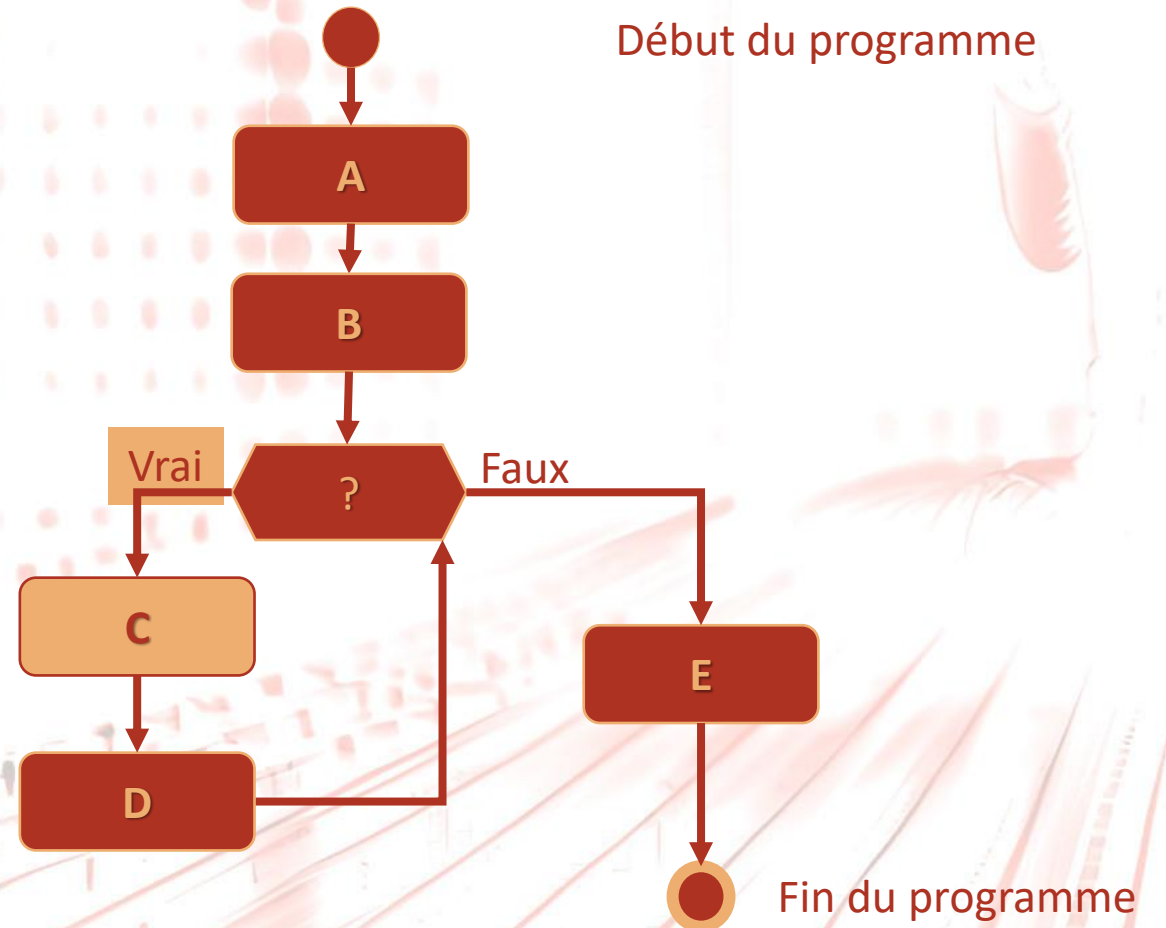
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 2
nb[2] vaut 0
i vaut 1
```



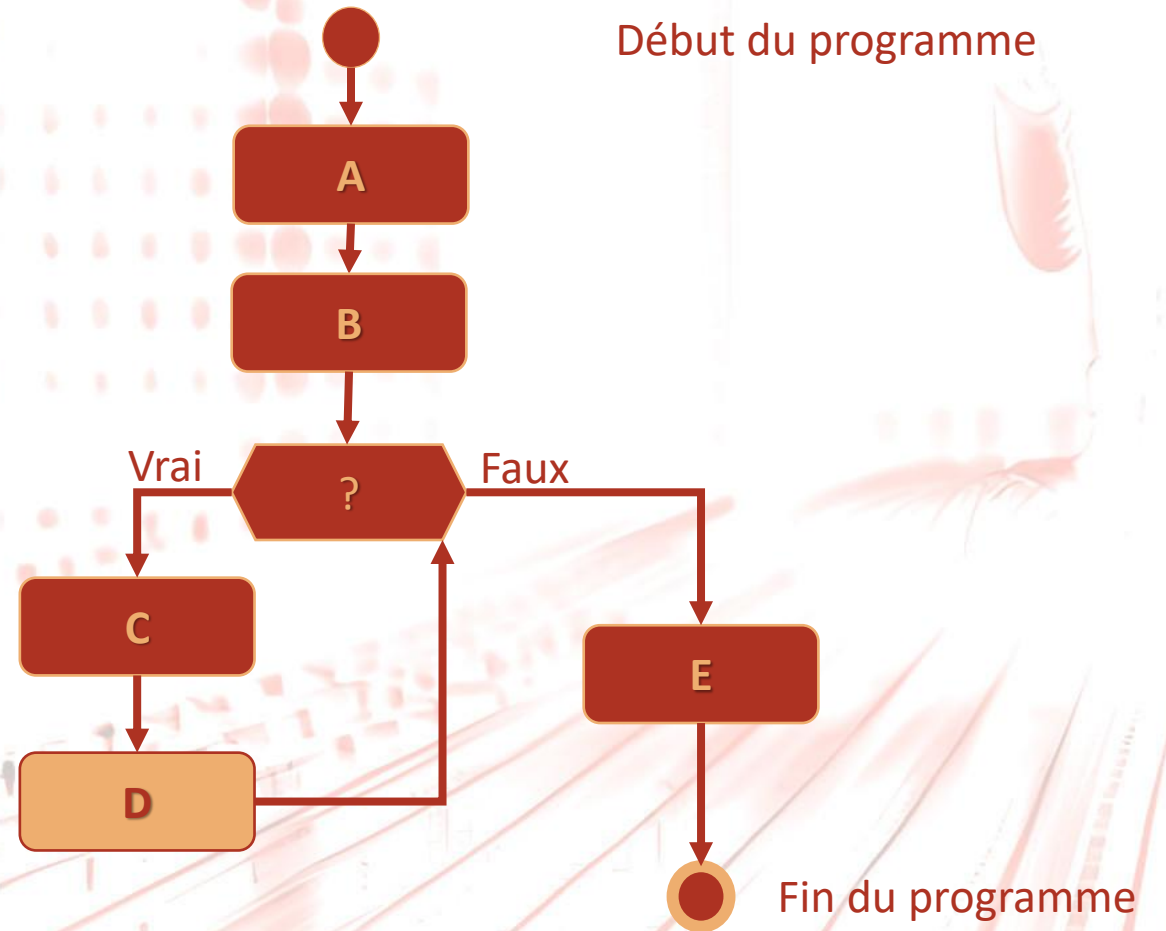
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 2
nb[2] vaut 0
i vaut 2
```



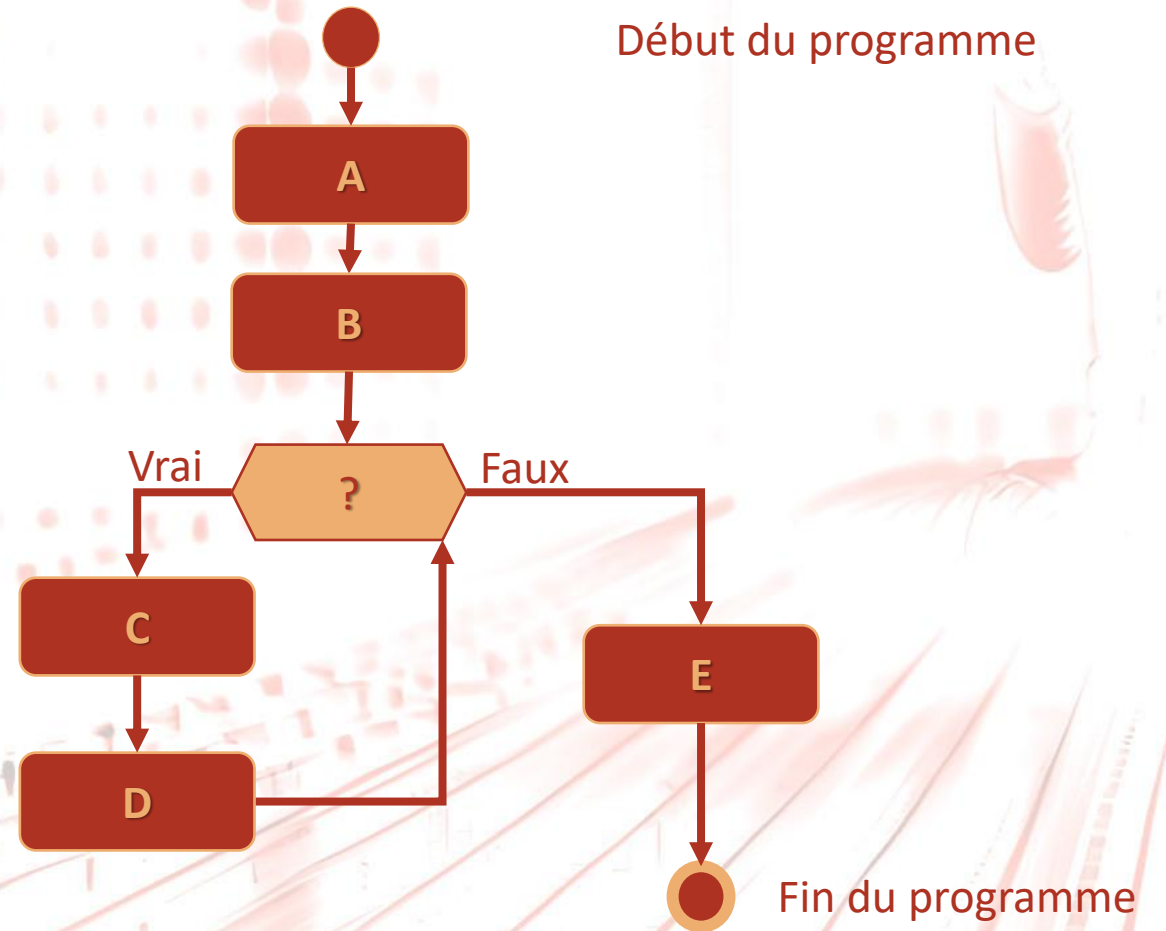
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0; i<3; i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 2
nb[2] vaut 0
i vaut 2
```



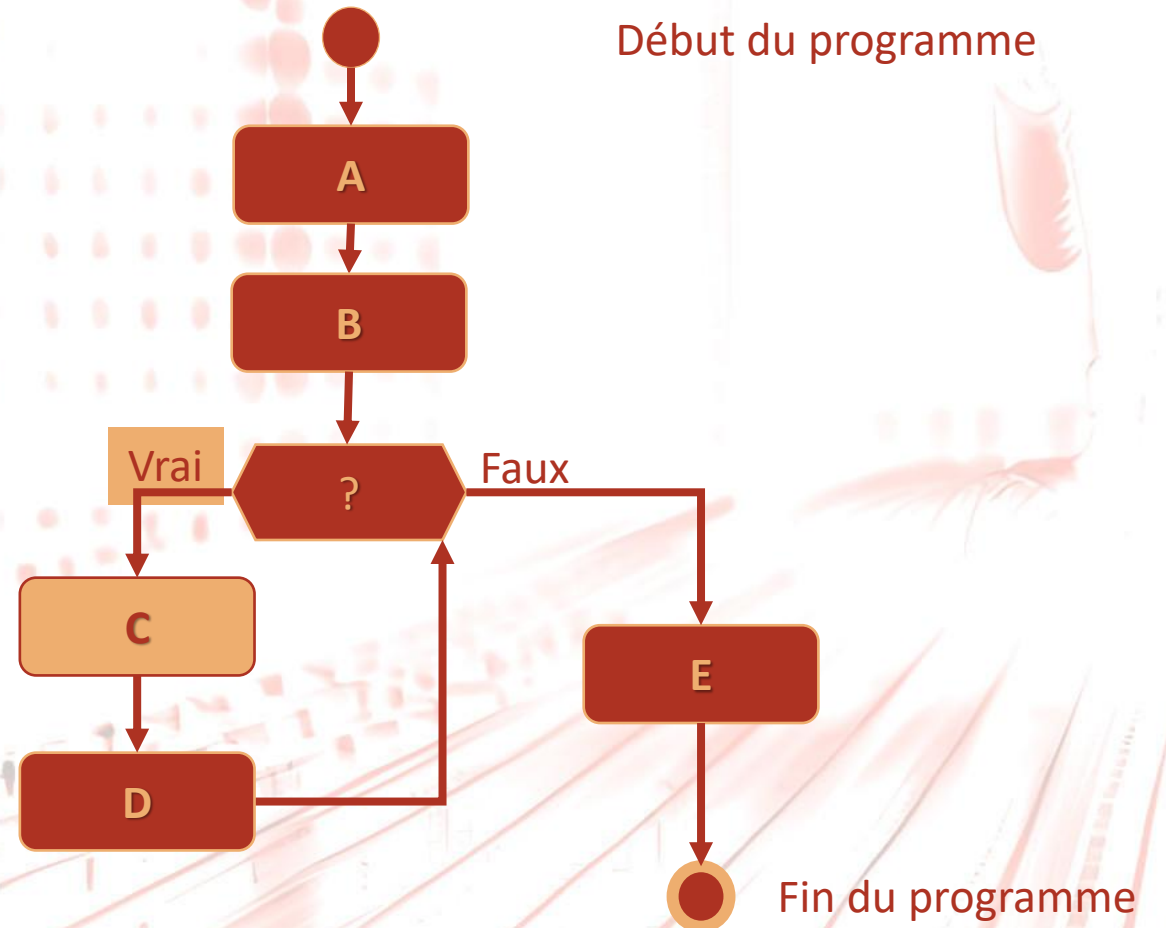
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}
```

nb[0] vaut 1
nb[1] vaut 2
nb[2] vaut 3
i vaut 2



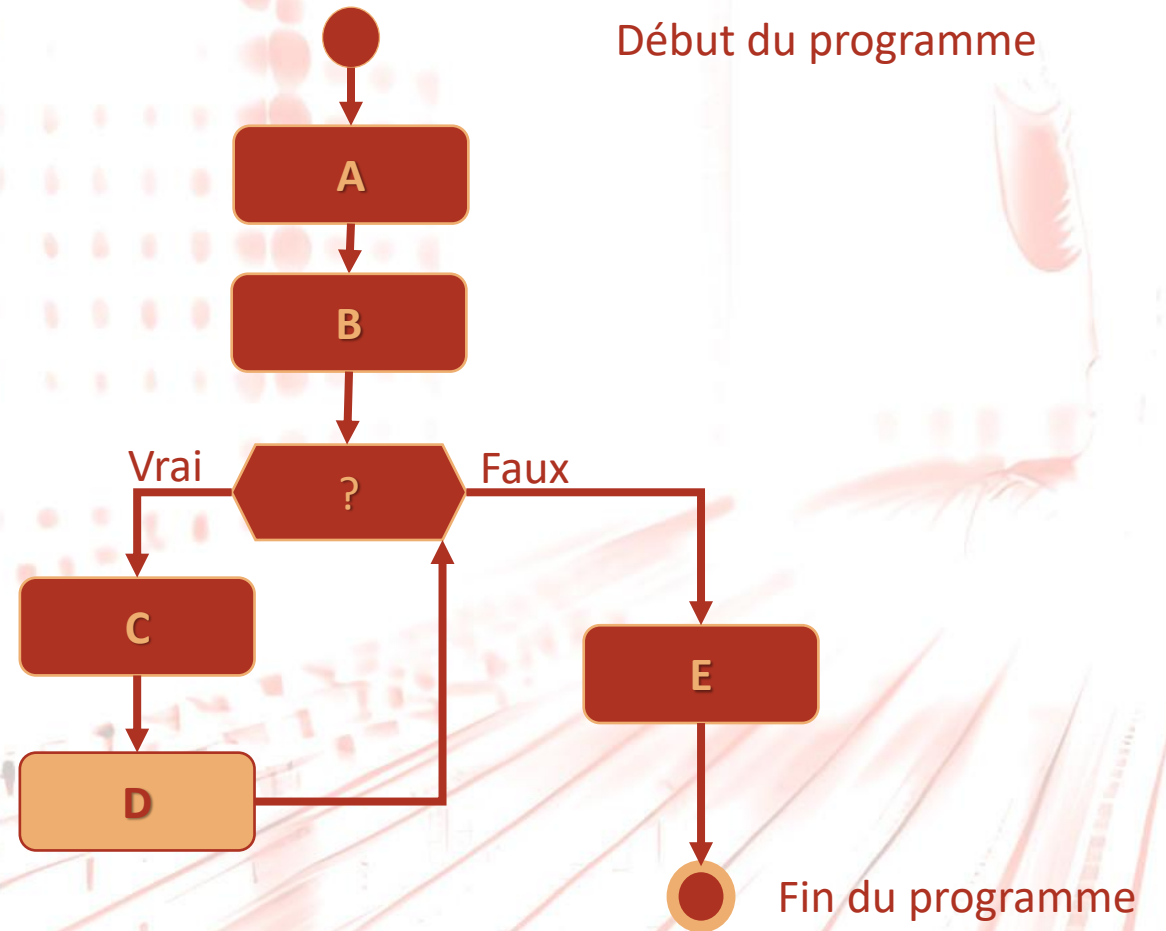
ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 2
nb[2] vaut 3
i vaut 3
```



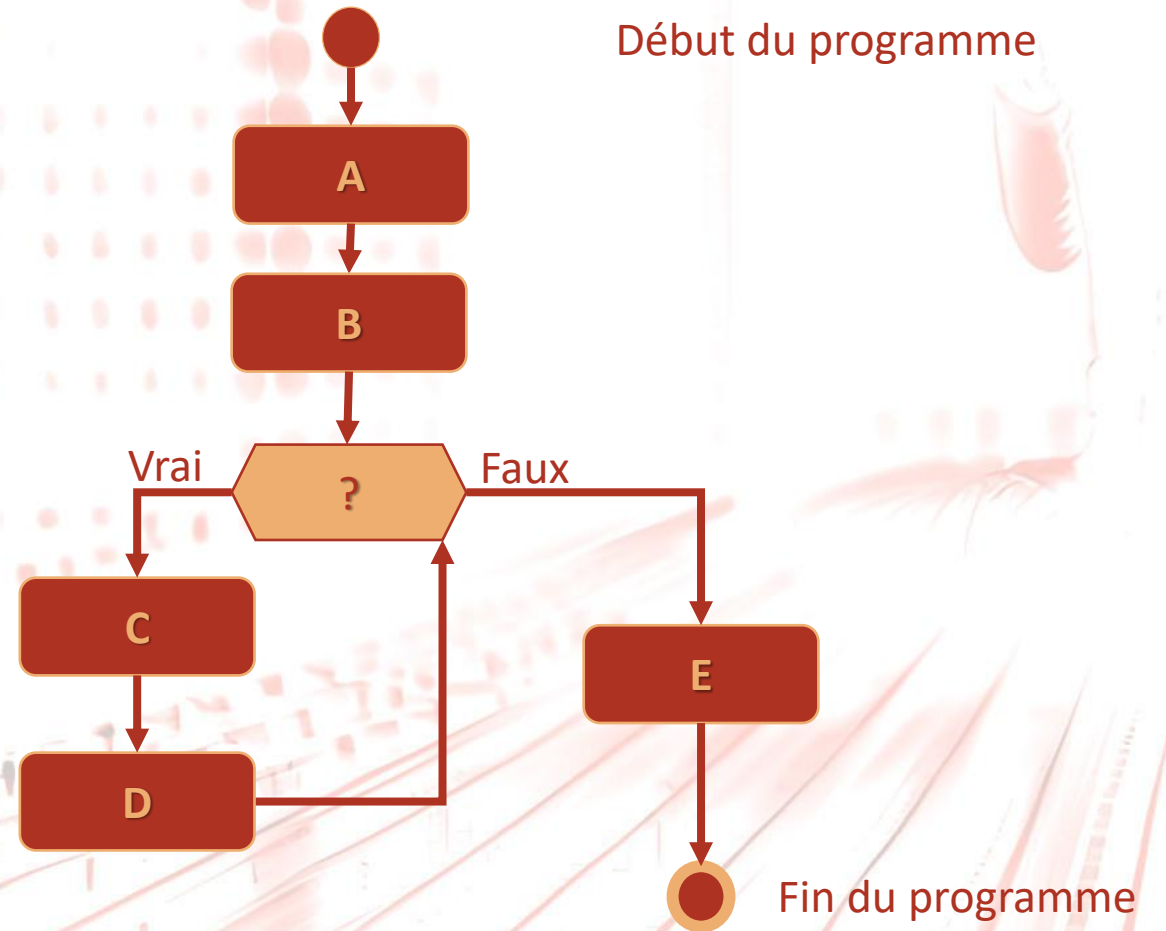
ITÉRATIONS: FOR

Programmation - Les structures

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#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0; i<3; i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 2
nb[2] vaut 3
i vaut 3
```



ITÉRATIONS: FOR

Programmation - Les structures

```
#include <iostream>
using namespace std;

int main(){
    int nb[3] = {0};
    for(int i=0;i<3;i++){
        nb[i]=i+1;
    }
    return 0;
}

nb[0] vaut 1
nb[1] vaut 2
nb[2] vaut 3
i vaut 3
```

